AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

CSC 2108: Introduction to Database

Fall 2023-2024

Section: F, Group: 01

Project Title

Tourism Management System

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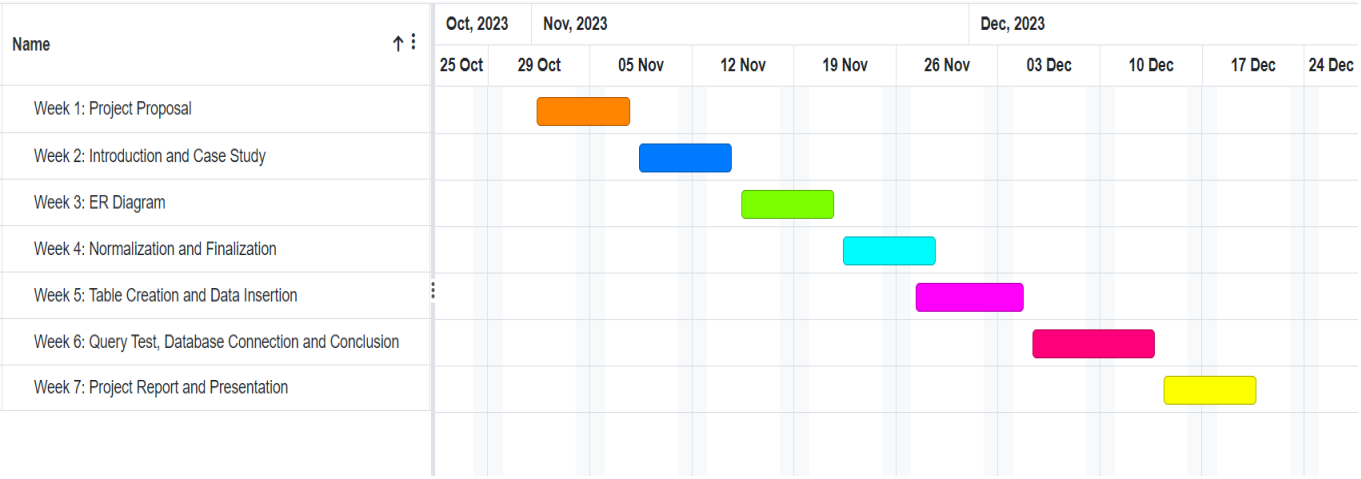
INTRODUCTION

Organizing travel plans has grown more difficult in the fast-paced world of today. Our project is based on a tourism management system that aims to establish a comprehensive and efficient solution for managing travel arrangements. This project aims to simplify the process of storing and retrieving important travel-related data, including reservations for flights, hotel booking, meals, and other accommodations, by building a centralized database. By successfully implementing this project, the travel industry can benefit from improved efficiency and seamless travel experience for both travel agencies and individual users.

GANTT CHART



CASE STUDY

In a tourism management system, a travel agency employs several employees to handle various tasks. Various types of employees work for a travel agency**. A travel agency** usually contains an agency ID, name, address, phone number, and email address. On the other hand, an **employee** has an employee ID, name, date of birth, job type, salary, and different skills. Customers can approach a travel agency. A travel agency services many customers. A **customer** registers with a customer ID, name, address, phone number, email, and membership level. All kinds of addresses are composed of city, country, and zip code. A customer makes payments for various facilities. A **payment** process may include Pay ID, amount, date, etc. A customer takes flights for traveling. A **flight** includes flight number, airline, departure and arrival which is composed by country and time. A customer uses various types of local transportation**.** Local **transportation** generally has a transport number, booking date, pick-up location, drop-in location, and mode of transport. Customers make reservations at hotels. On the other hand, A **hotel** offers accommodations to the customers and the customer gets accommodations. To make **accommodations**, there is an accommodation ID, in date, out date, room type and room number. A hotel manages food booking. Different types of food bookings are offered by a hotel. **Food booking** is maintained by booking ID, booking date, meal type, and total number of guests. **In total, there are 9 entities in the project.**

ER DIAGRAM

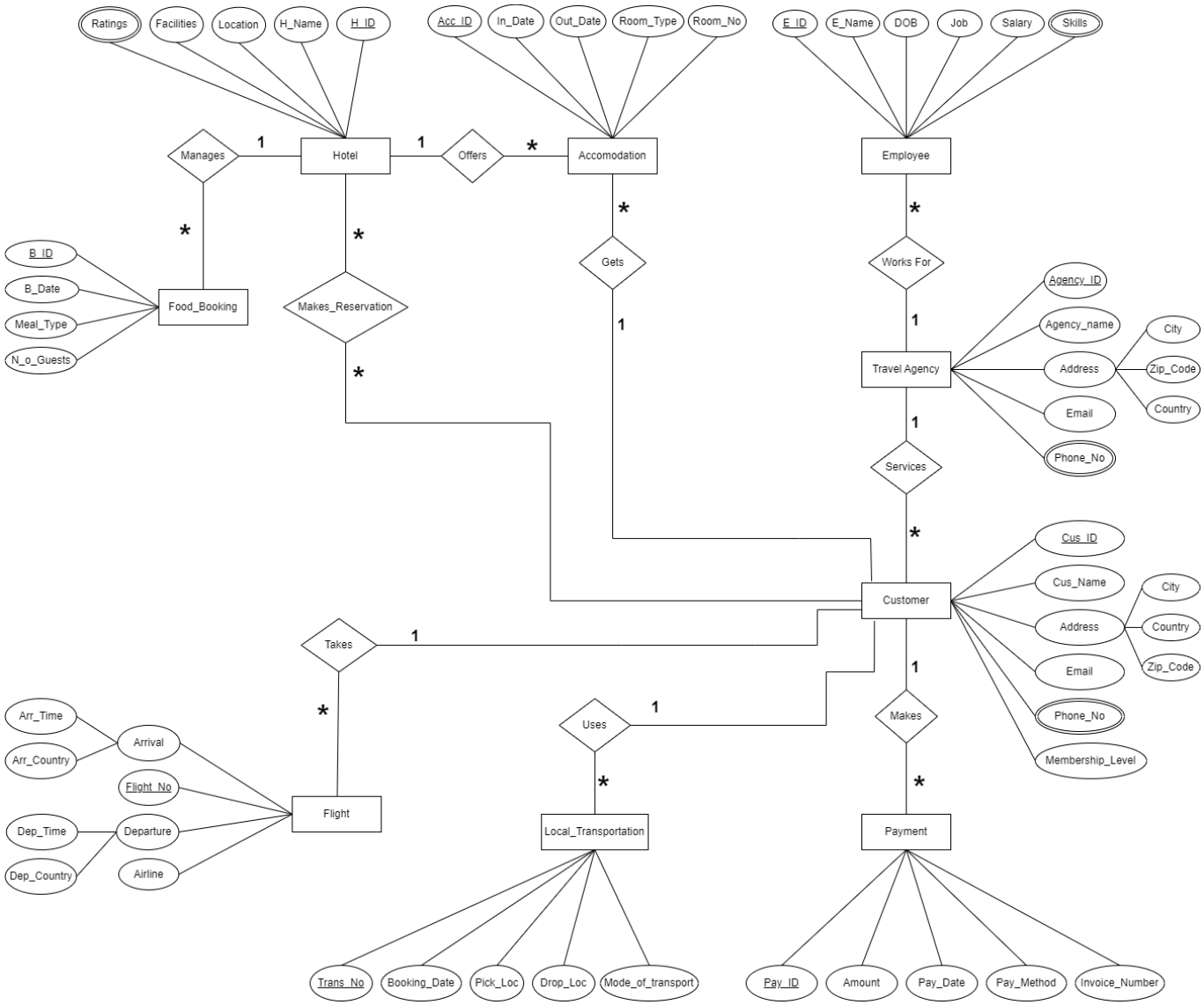


Fig-1: ER-Diagram for Tourism Management System

NORMALIZATION

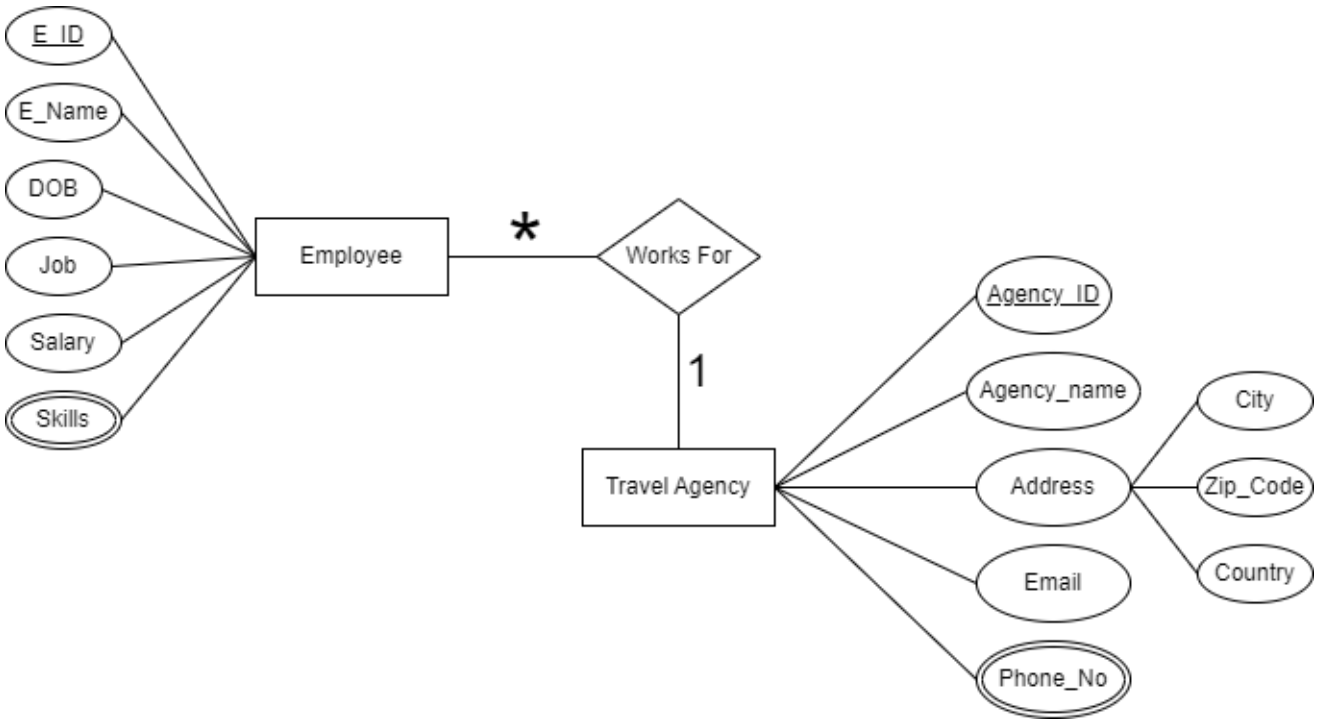


Fig-2: ER-Diagram for Employee – works for – Travel Agency

Relation: Many to One

UNF: Agency ID, Agency_Name, City, Country, Zip_code, Phone_No, Email, E_ID, E_Name, DOB, Job Title, Salary, Skills

1NF: Multivalued attributes: Phone_No, Skills.
Agency ID, Agency_Name, City, Country, Zip_code, Email, E_ID, E_Name, Date of Birth, Job Title, Salary

2NF: 1. Agency ID, Phone_No
1. E_ID, Skills
2. Agency ID, Agency_Name, City, Country, Zip_code, Email
3. E_ID, E_Name, DOB, Job Title, Salary, **Agency ID**

3NF: 1. Agency ID, City, Country, Zip_code
2. Agency ID, Phone_No
3. E_ID, Skills
4. Agency ID, Agency_Name, Email
5. E_ID, E_Name, DOB, Job Title, Salary, **Agency ID**

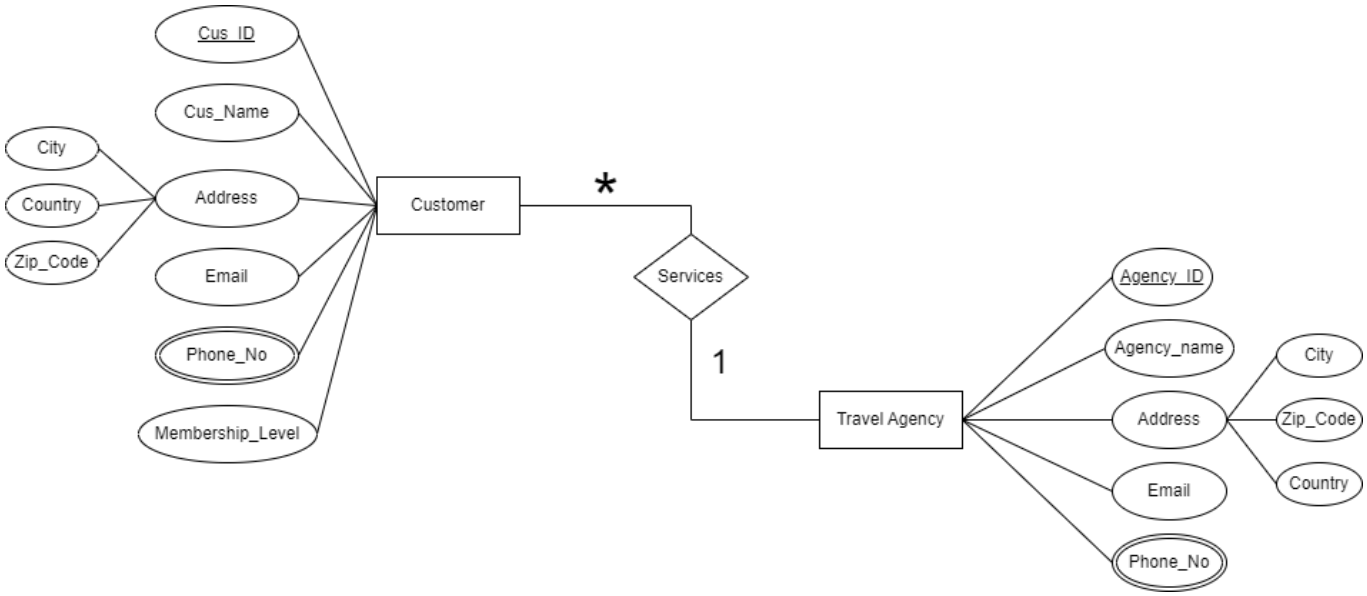


Fig-3: ER-Diagram for Travel Agency – services – Customer

Relation: One to Many

UNF: Agency ID, Agency_Name, City, Country, Zip_code, Phone_No, Email, Cus_ID, Cus_Name, City, Country, Zip_code, Email, Phone_No, Membership_Level

1NF: Multivalued attributes: Phone_No, Phone_No.
Agency ID, Agency_Name, City, Country, Zip_code, Email, Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level

2NF:
1. Agency ID, Phone_No
2. Cus_ID, Phone_No
3. Agency ID, Agency_Name, City, Country, Zip_code, Email
4. Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level, **Agency ID**

3NF:
1. Agency ID, City, Country, Zip_code
2. Cus_ID, City, Country, Zip_code
3. Agency ID, Phone_No
4. Cus_ID, Phone_No
5. Agency ID, Agency_Name, Email
6. Cus_ID, Cus_Name, Email, Membership_Level, **Agency ID**

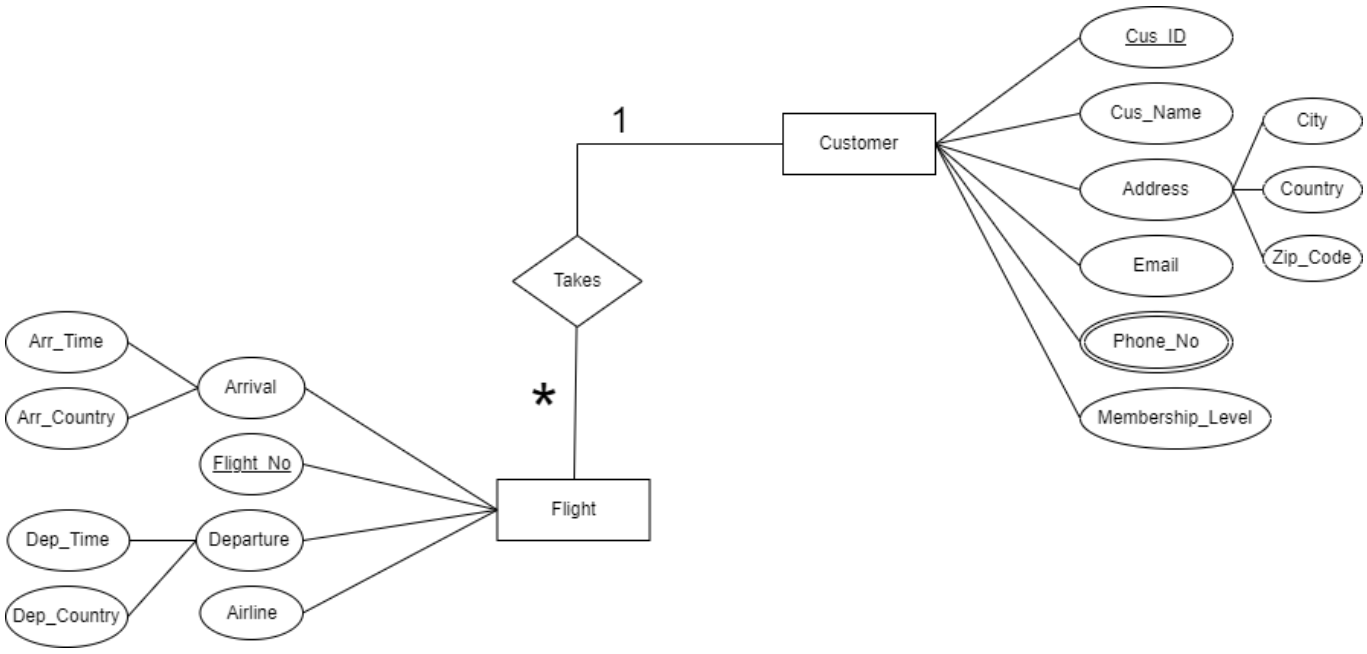


Fig-4: ER-Diagram for Customer – takes – Flight

Relation: One to Many

UNF: Cus_ID, Cus_Name, City, Country, Zip_code, Email, phone_no, Membership_Level, Flight_No, Airline, Dep_Country, Arr_Country, Dep_Time, Arr_Time

1NF: Multivalued attributes: Phone_No
Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level, Flight_No, Airline, Dep_Country, Arr_Country, Dep_Time, Arr_Time

2NF: 1. Cus_ID, Phone_No
2.Flight_No, Airline, Dep_Country, Arr_Country, Dep_Time, Arr_Time, Cus_ID
3.Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level

3NF: 1. Cus_ID, Phone_No

2. Flight_No, Dep_Country, Arr_Country, Dep_Time, Arr_Time
3. Cus_ID, City, Country, Zip_code
4. Cus_ID, Cus_Name, Email, Membership_Level
5. Flight_No, Airline, **Cus_ID**

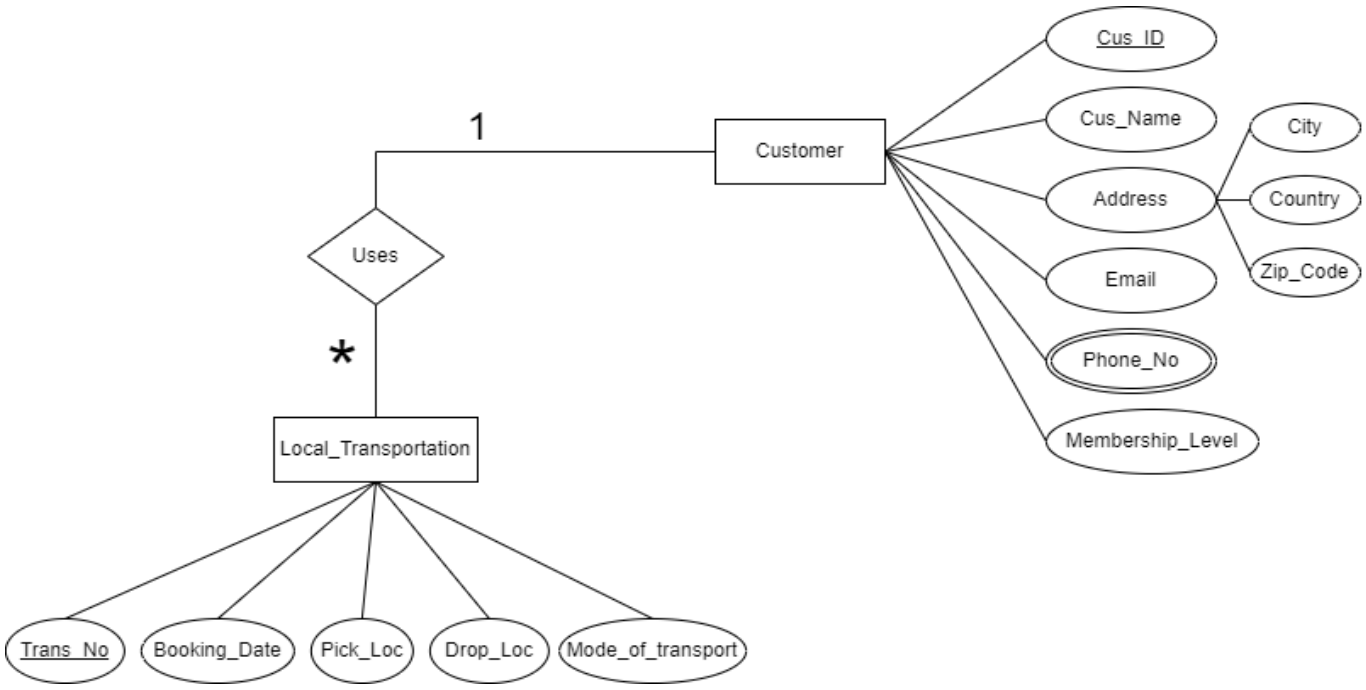


Fig-5: ER-Diagram for Customer – uses – Local_Transportation

Relation: One to Many

UNF: Cus_ID, Cus_Name, City, Country, Zip_code, Email, Phone_No, Membership_Level, Trans_No, Booking_Date, Pick_Loc, Drop_Loc, Mode_of_Transport

1NF: Multivalued attributes: Phone_No
Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level, Trans_No, Booking_Date, Pick_Loc, Drop_Loc, Mode_of_Transport

2NF: 1. Cus_ID, Phone_No

2.Trans_No, Booking_Date, Pick_Loc, Drop_Loc, Mode_of_Transport, **Cus_ID**
3.Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level

3NF: 1. Cus_ID, City, Country, Zip_code
2. Cus_ID, Phone_No
3. Trans_No, Booking_Date, Pick_Loc, Drop_Loc, Mode of Transport, **Cus_ID**
4. Cus_ID, Cus_Name, Email, Membership_Level

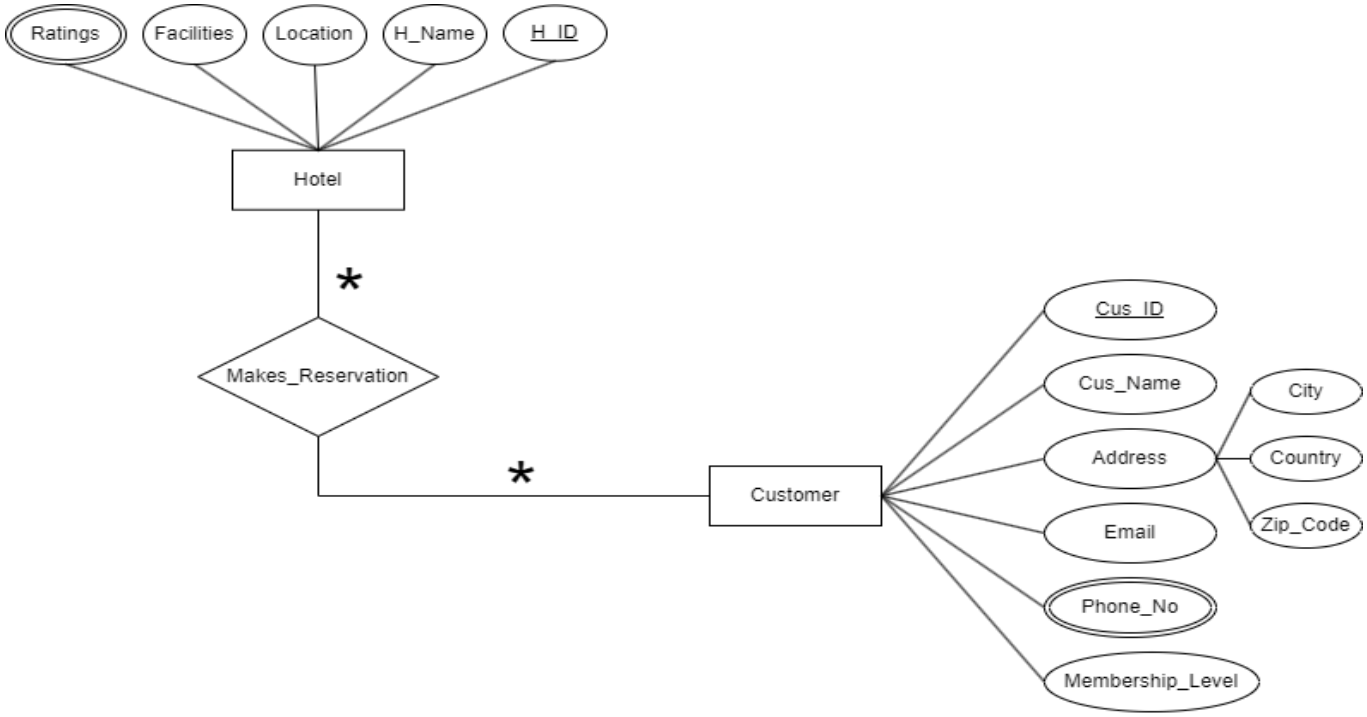


Fig-6: ER-Diagram for Customer – makes_reservation – Hotel

Relation: Many to Many

UNF: Cus_ID, Cus_Name, City, Country, Zip_code, Email, Phone_No, Membership_Level, H_ID, H_Name, Location, Facilites, Ratings

1NF: Multivalued attributes: Phone_No, Ratings
Cus_ID, Cus_Name, City, Country, Zip_code, Email, Phone_No, Membership_Level, H_ID, H_Name, Location, Facilites

2NF: 1. Cus_ID, Phone_No

2.H_ID, Ratings
3.Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level
4.H_ID, H_Name, Location, Facilities, **Cus_ID**

3NF: 1. Cus_ID, City, Country, Zip_code
2. Cus_ID, Phone_No
3. H_ID, Ratings
4. Cus_ID, Cus_Name, Email, Membership_Level
5. H_ID, H_Name, Location, Facilities, **Cus_ID**

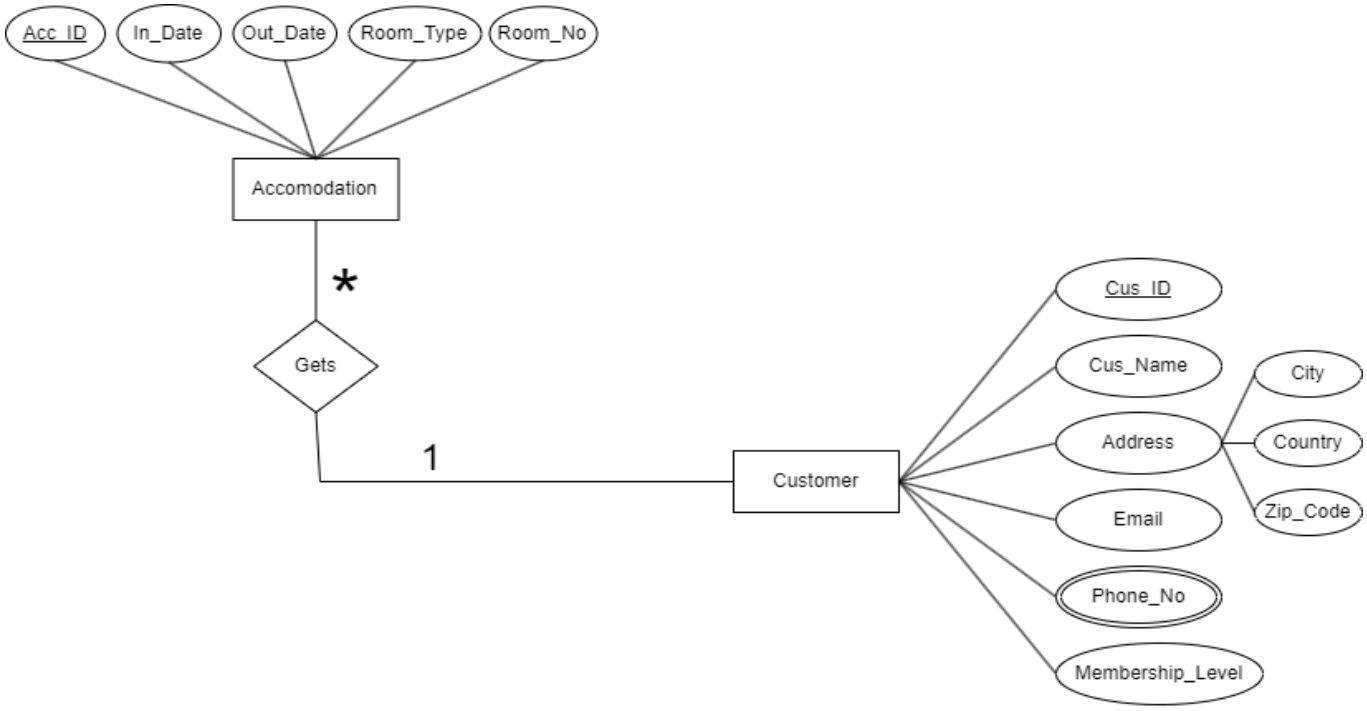


Fig-7: ER-Diagram for Customer – gets – Accommodation

Relation: One to Many

UNF: Cus_ID, Cus_Name, City, Country, Zip_code, Email, phone_no, Membership_Level, Acc_ID, In_Date, Out_Date, Room_Type, Room_No

1NF: Multivalued attributes: Phone_No
Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level, Acc_ID, In_Date, Out_Date, Room_Type, Room_No

2NF: 1. Cus_ID, Phone_No

2.Acc_ID, In_Date, Out_Date, Room_Type, Room_No, **Cus_ID**
3.Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level

3NF: 1. Cus_ID, City, Country, Zip_code
2. Cus_ID, Phone_No
3. Cus_ID, Cus_Name, Email, Membership_Level
4. Acc_ID, In_Date, Out_Date, Room_Type, Room_No, **Cus_ID**

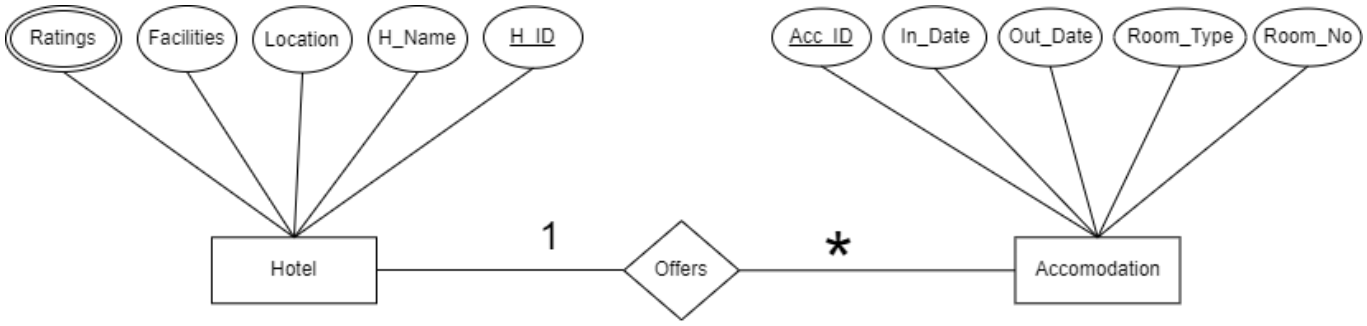


Fig-8: ER-Diagram for Hotel – offers – Accommodation

Relation: One to Many

UNF: Acc_ID, In_Date, Out_Date, Room_Type, Room_No, H_ID, H_name, location, Facilities, Ratings

1NF: Multivalued attributes: Ratings
Acc_ID, In_Date, Out_Date, Room_Type, Room_No, H_ID, H_name, location, Facilites

2NF: 1. H_ID, Ratings

2.Acc_ID, In_Date, Out_Date, Room_Type, Room_No, **H_ID**
3.H_ID, H_name, location, Facilites

3NF: 1. H_ID, Ratings

2. Acc_ID, In_Date, Out_Date, Room_Type, Room_No, **H_ID**
3. H_ID, H_name, location, Facilites

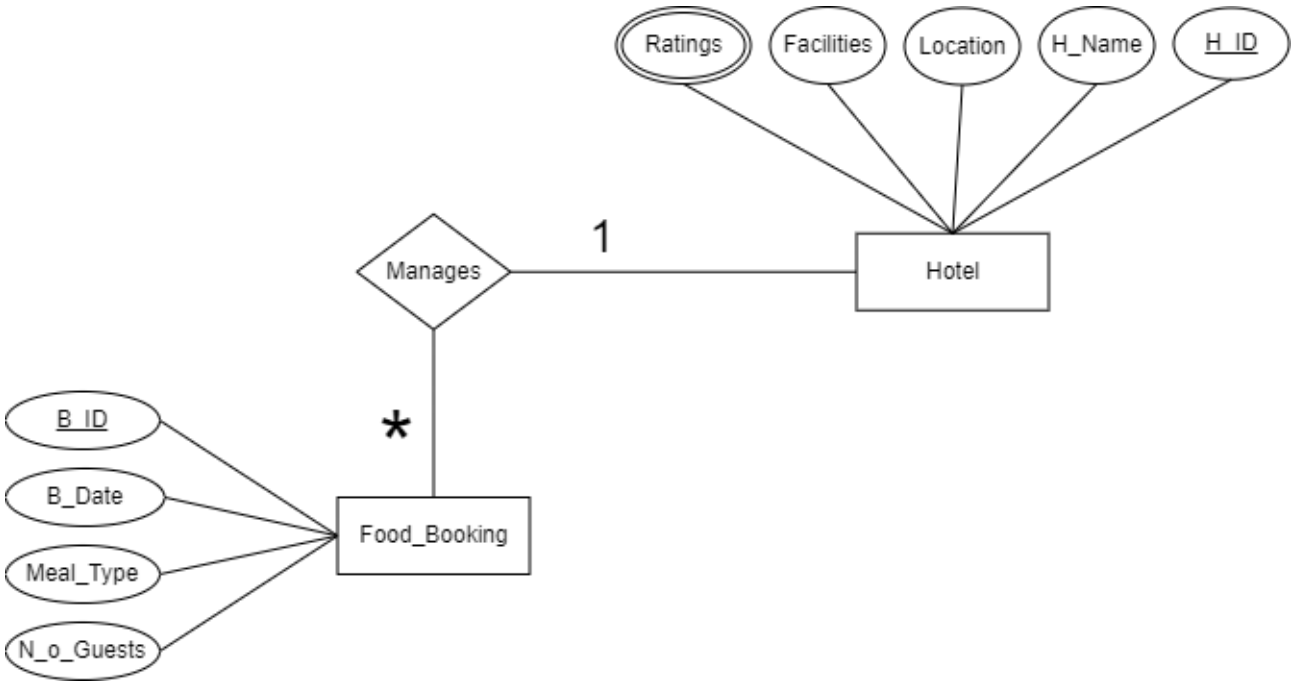


Fig-9: ER-Diagram for Hotel – manages – Food_Booking

Relation: One to Many

UNF: H_ID, H_name, location, Facilities, Ratings, B_ID, B_Date, Meal_Type, N_o_Guests

1NF: Multivalued attributes: Ratings
H_ID, H_name, location, Facilities, B_ID, B_Date, Meal_Type, N_o_Guests

2NF: 1. H_ID, Ratings

2.H_ID, H_name, location, Facilities
3.B_ID, B_Date, Meal_Type, N_o_Guests, **H_ID**

3NF: 1. H_ID, Ratings

2. H_ID, H_name, location, Facilities
3. B_ID, B_Date, Meal_Type, N_o_Guests, **H_ID**

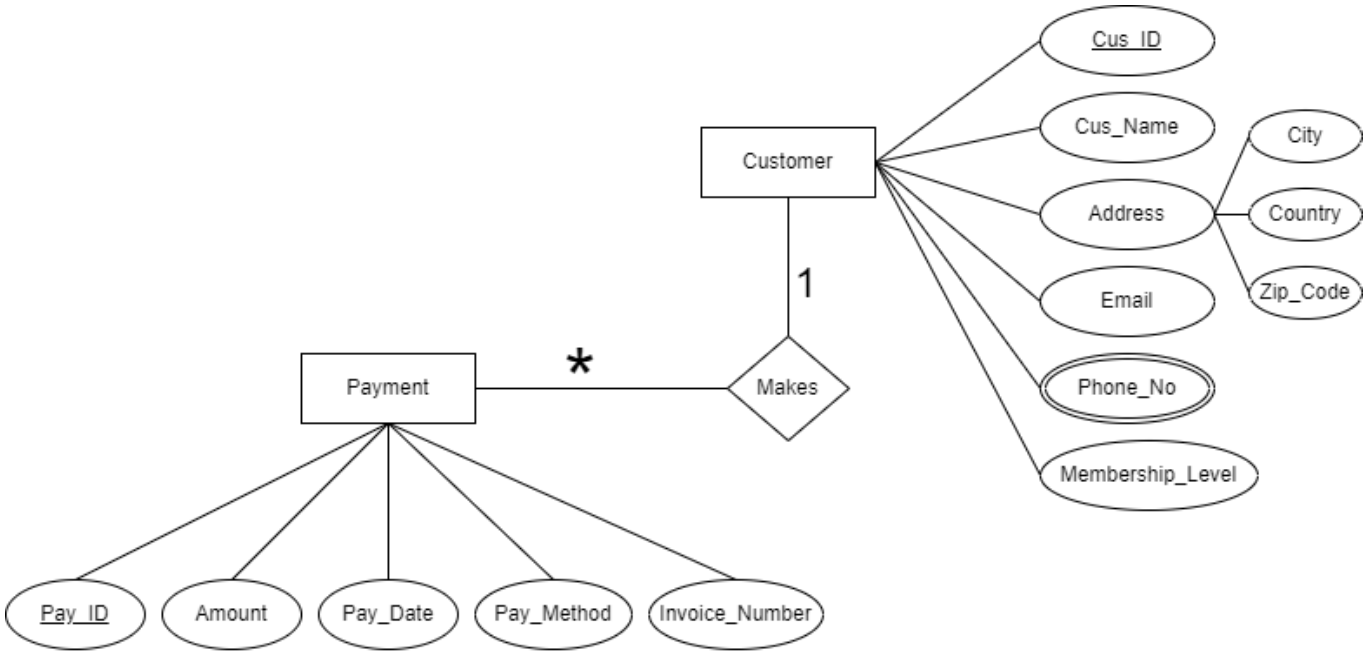


Fig-10: ER-Diagram for Customer – makes – Payment

Relation: One to Many

UNF: Cus_ID, Cus_Name, City, Country, Zip_code, Email, phone_no, Membership_Level, Pay_ID, Amount, Pay_Date, Pay_Method, Invoice_Number, Phone_No

1NF: Multivalued attributes: Phone_No
Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level, Pay_ID, Amount, Pay_Date, Pay_Method, Invoice_Number_Phone_No

2NF: 1. Cus_ID, Phone_No

2.Cus_ID, Cus_Name, City, Country, Zip_code, Email, Membership_Level
3.Pay_ID, Amount, Pay_Date, Pay_Method, Invoice_Number, **Cus_ID**

3NF: 1. Cus_ID, City, Country, Zip_code
2. Cus_ID, Phone_No
3. Cus_ID, Cus_Name, Email, Membership_Level
4. Pay_ID, Amount, Pay_Date, Pay_Method, Invoice_Number, **Cus_ID**

FINALIZATION

1. Agency ID, City, Country, Zip_code
2. Agency ID, Phone_No
3. E_ID, Skills
4. Agency ID, Agency_Name, Email
5. E_ID, E_Name, DOB, Job Title, Salary, **Agency ID**
~~**6.** Agency ID, City, Country, Zip_code~~
7. Cus_ID, City, Country, Zip_code
~~**8.** Agency ID, Phone_No~~
9. Cus_ID, Phone_No
~~**10.** Agency ID, Agency_Name, Email~~
11. Cus_ID, Cus_Name, Email, Membership_Level, **Agency ID**
~~**12.** Cus_ID, City, Country, Zip_code~~
~~**13.** Cus_ID, Phone_No~~
14. Trans_No, Booking_Date, Pick_Loc, Drop_Loc, Mode of Transport, Cus_ID
~~**15.** Cus_ID, Cus_Name, Email, Membership_Level,**Agency ID**~~
~~**16.** Cus_ID, City, Country, Zip_code~~
~~**17.** Cus_ID, Phone_No~~
18. H_ID, Ratings
~~**19.** Cus_ID, Cus_Name, Email, Membership_Level~~
20. H_ID, H_Name, Location, Facilities**, Cus_ID**
~~**21.** Cus_ID, City, Country, Zip_code~~
~~**22.** Cus_ID, phone_no~~
~~**23.** Cus_ID, Cus_Name, Email, Membership_Level~~
24. Pay_ID, Amount, Pay_Date, Pay_Method, Invoice_Number, Cus_ID
~~**25.** H_ID, Ratings~~
~~**26.** H_ID, H_name, location, Facilities ,**Cus_id**~~
27. B_ID, B_Date, Meal_Type, N_o_Guests, **H_ID**
~~**28.** Cus_ID, City, Country, Zip_code~~
~~**29.** Cus_ID, phone_no~~
~~**30.** Cus_ID, Cus_Name, Email, Membership_Level~~
31. Acc_ID, In_Date, Out_Date, Room_Type, Room_No, **Cus_ID**
~~**32.** H_ID, Ratings~~
~~**33.** Acc_ID, In_Date, Out_Date, Room_Type, Room_No, **Cus_ID**~~
~~**34.** H_ID, H_name, location, Facilities~~
~~**35.** Cus_ID, Phone_No~~
36. Flight_No, Dep_Country, Arr_Country, Dep_Time, Arr_Time
~~**37.** Cus_ID, City, Country, Zip_code~~
~~**38.** Cus_ID, Cus_name,Email, Membership_Level~~
39. Filght_no,Airline,**Cus_id**

TABLE CREATION

1. AGENCY_INFO:

Autocommit Display 10

```
CREATE TABLE AGENCY_INFO
(AGENCY_ID VARCHAR2(20) PRIMARY KEY,
AGENCY_NAME VARCHAR2(20) NOT NULL,
EMAIL VARCHAR2(20) UNIQUE);

DESC AGENCY_INFO;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **AGENCY_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
AGENCY_INFO	AGENCY_ID	Varchar2	20	-	-	1	-	-	-
	AGENCY_NAME	Varchar2	20	-	-	-	-	-	-
	EMAIL	Varchar2	20	-	-	-	✓	-	-

1 - 3

2. AGENCY_LOC:

Autocommit Display 10

```
CREATE TABLE AGENCY_LOC
(AGENCY_ID VARCHAR2(20) PRIMARY KEY,
CITY VARCHAR2(20) NOT NULL,
COUNTRY VARCHAR2(20) NOT NULL,
ZIP_CODE NUMBER NOT NULL);

DESC AGENCY_LOC;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **AGENCY_LOC**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
AGENCY_LOC	AGENCY_ID	Varchar2	20	-	-	1	-	-	-
	CITY	Varchar2	20	-	-	-	-	-	-
	COUNTRY	Varchar2	20	-	-	-	-	-	-
	ZIP_CODE	Number	-	-	-	-	-	-	-

1 - 4

3. AGENCY_CON:

Autocommit Display 10

```
CREATE TYPE MULTIVALUE_TYPE AS TABLE OF VARCHAR2(15);
CREATE TABLE AGENCY_CON
( AGENCY_ID VARCHAR2(20) PRIMARY KEY,
PHONE_NO MULTIVALUE_TYPE)

NESTED TABLE PHONE_NO STORE AS PHONE_NUMBERS_TABLE;
DESC AGENCY_CON;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **AGENCY_CON**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
AGENCY_CON	AGENCY_ID	Varchar2	20	-	-	1	-	-	-
	PHONE_NO	Multivalue_Type	16	-	-	-	✓	-	-

1 - 2

4. EMP_INFO:

Autocommit Display 10

```
CREATE TABLE EMP_INFO
(E_ID VARCHAR2(20) PRIMARY KEY,
E_NAME VARCHAR2(20) NOT NULL,DOB DATE NOT NULL,
JOB_TITLE VARCHAR2(20) NOT NULL,SALARY NUMBER(20) NOT NULL,
AGENCY_ID VARCHAR2(20) NOT NULL );
ALTER TABLE EMP_INFO
ADD FOREIGN KEY (AGENCY_ID) REFERENCES AGENCY_INFO (AGENCY_ID);
DESC EMP_INFO;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **EMP_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP_INFO	E_ID	Varchar2	20	-	-	1	-	-	-
	E_NAME	Varchar2	20	-	-	-	-	-	-
	DOB	Date	7	-	-	-	-	-	-
	JOB_TITLE	Varchar2	20	-	-	-	-	-	-
	SALARY	Number	-	20	0	-	-	-	-
	AGENCY_ID	Varchar2	20	-	-	-	-	-	-

1 - 6

5. EMP_SKILL:

Autocommit Display 10

```
CREATE TABLE EMP_SKILL
(E_ID VARCHAR2(20) PRIMARY KEY,
SKILLS MULTIVALUE_TYPE)
NESTED TABLE SKILLS STORE AS PHONE_NUMBERS_TABLE4;

DESC EMP_SKILL;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **EMP_SKILL**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP_SKILL	E_ID	Varchar2	20	-	-	1	-	-	-
	SKILLS	Multivalue_Type	16	-	-	-	✓	-	-

1 - 2

6. CUS_INFO:

Autocommit Display 10

```
CREATE TABLE CUS_INFO
(CUS_ID VARCHAR2(20) PRIMARY KEY,
CUS_NAME VARCHAR2(100) NOT NULL, EMAIL VARCHAR2(100) UNIQUE,
MEMBERSHIP_LEVEL VARCHAR2(20), AGENCY_ID VARCHAR2(20));
ALTER TABLE CUS_INFO ADD CONSTRAINT
AGENCY FOREIGN KEY (AGENCY_ID) REFERENCES AGENCY_INFO (AGENCY_ID);

DESC CUS_INFO;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **CUS_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CUS_INFO	CUS_ID	Varchar2	20	-	-	1	-	-	-
	CUS_NAME	Varchar2	100	-	-	-	-	-	-
	EMAIL	Varchar2	100	-	-	-	✓	-	-
	MEMBERSHIP_LEVEL	Varchar2	20	-	-	-	✓	-	-
	AGENCY_ID	Varchar2	20	-	-	-	✓	-	-

1 - 5

7. CUS_LOC:

Autocommit Display 10

```
CREATE TABLE CUS_LOC
(CUS_ID VARCHAR2(20) PRIMARY KEY,
COUNTRY VARCHAR2(20) NOT NULL, CITY VARCHAR2(20) NOT NULL,
ZIP_CODE VARCHAR2(20));

DESC CUS_LOC;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **CUS_LOC**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CUS_LOC	CUS_ID	Varchar2	20	-	-	1	-	-	-
	COUNTRY	Varchar2	20	-	-	-	-	-	-
	CITY	Varchar2	20	-	-	-	-	-	-
	ZIP_CODE	Varchar2	20	-	-	-	✓	-	-

1 - 4

8.CUS_CON:

Autocommit Display 10

```
CREATE TABLE CUS_CON
(CUS_ID VARCHAR(20) PRIMARY KEY,
PHONE_NO MULTIVALUE_TYPE)
NESTED TABLE PHONE_NO STORE AS PHONE_NUMBERS_TABLE1;

DESC CUS_CON;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **CUS_CON**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CUS_CON	CUS_ID	Varchar2	20	-	-	1	-	-	-
	PHONE_NO	Multivalue_Type	16	-	-	-	✓	-	-

1 - 2

9.AIRLINE:

Autocommit Display 10

```
CREATE TABLE AIRLINE
(FLIGHT_NO VARCHAR2(20) PRIMARY KEY,
AIRLINE VARCHAR2(20) NOT NULL,CUS_ID VARCHAR2(20) NOT NULL);
ALTER TABLE AIRLINE ADD FOREIGN KEY (CUS_ID) REFERENCES CUS_INFO(CUS_ID);

DESC AIRLINE;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **AIRLINE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
AIRLINE	FLIGHT_NO	Varchar2	20	-	-	1	-	-	-
	AIRLINE	Varchar2	20	-	-	-	-	-	-
	CUS_ID	Varchar2	20	-	-	-	-	-	-

1 - 3

10. AIRLINE_INFO:

```
CREATE TABLE AIRLINE_INFO
(FLIGHT_NO VARCHAR2(20) PRIMARY KEY,
DEP_COUNTRY VARCHAR2(20) NOT NULL,ARR_COUNTRY VARCHAR2(20) NOT NULL,
DEP_TIME VARCHAR(20) NOT NULL,ARR_TIME VARCHAR2(20) NOT NULL);

DESC AIRLINE_INFO;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **AIRLINE_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
AIRLINE_INFO	FLIGHT_NO	Varchar2	20	-	-	1	-	-	-
	DEP_COUNTRY	Varchar2	20	-	-	-	-	-	-
	ARR_COUNTRY	Varchar2	20	-	-	-	-	-	-
	DEP_TIME	Varchar2	20	-	-	-	-	-	-
	ARR_TIME	Varchar2	20	-	-	-	-	-	-

1 - 5

11. TRANSPORTATION:

```
CREATE TABLE TRANSPORTATION (TRANS_NO VARCHAR(20) PRIMARY KEY,
BOOKING_DATE DATE UNIQUE, PICK_LOC VARCHAR(20) NOT NULL,
DROP_LOC VARCHAR(20) NOT NULL, MODE_OF_TRANSPORT VARCHAR(10), CUS_ID VARCHAR2(20));
ALTER TABLE TRANSPORTATION
ADD CONSTRAINT CUSTOMERS FOREIGN KEY (CUS_ID) REFERENCES CUS_INFO (CUS_ID);

DESC TRANSPORTATION;
```

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **TRANSPORTATION**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TRANSPORTATION	TRANS_NO	Varchar2	20	-	-	1	-	-	-
	BOOKING_DATE	Date	7	-	-	-	✓	-	-
	PICK_LOC	Varchar2	20	-	-	-	-	-	-
	DROP_LOC	Varchar2	20	-	-	-	-	-	-
	MODE_OF_TRANSPORT	Varchar2	10	-	-	-	✓	-	-
	CUS_ID	Varchar2	20	-	-	-	✓	-	-

1 - 6

12.HOTEL_INFO:

```
CREATE TABLE HOTEL_INFO(H_ID VARCHAR2(20) PRIMARY KEY,
H_NAME VARCHAR2(20) NOT NULL,LOCATION VARCHAR2(20) NOT NULL,
FACILITES VARCHAR2(20) NOT NULL,CUS_ID VARCHAR2(20) NOT NULL);
ALTER TABLE HOTEL_INFO
ADD FOREIGN KEY (CUS_ID) REFERENCES CUS_INFO(CUS_ID);

DESC HOTEL_INFO;
```

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **HOTEL_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
HOTEL_INFO	H_ID	Varchar2	20	-	-	1	-	-	-
	H_NAME	Varchar2	20	-	-	-	-	-	-
	LOCATION	Varchar2	20	-	-	-	-	-	-
	FACILITES	Varchar2	20	-	-	-	-	-	-
	CUS_ID	Varchar2	20	-	-	-	-	-	-

1 - 5

13.ACCOMODATION:

```
CREATE TABLE ACCOMMODATION(ACC_ID VARCHAR2(20) PRIMARY KEY,
IN_DATE DATE NOT NULL, OUT_DATE DATE NOT NULL, ROOM_TYPE VARCHAR2(20),
ROOM_NO NUMBER(7,0) UNIQUE, CUS_ID VARCHAR2(20), H_ID VARCHAR2(20));
ALTER TABLE ACCOMMODATION
ADD CONSTRAINT CUSTOMER1 FOREIGN KEY (CUS_ID) REFERENCES CUS_INFO;
ALTER TABLE ACCOMMODATION
ADD CONSTRAINT HOTEL FOREIGN KEY (H_ID) REFERENCES HOTEL_INFO;

DESC ACCOMMODATION;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **ACCOMMODATION**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ACCOMMODATION	ACC_ID	Varchar2	20	-	-	1	-	-	-
	IN_DATE	Date	7	-	-	-	-	-	-
	OUT_DATE	Date	7	-	-	-	-	-	-
	ROOM_TYPE	Varchar2	20	-	-	-	✓	-	-
	ROOM_NO	Number	-	7	0	-	✓	-	-
	CUS_ID	Varchar2	20	-	-	-	✓	-	-
	H_ID	Varchar2	20	-	-	-	✓	-	-

1 - 7

14. FOOD_BOOKING:

```
CREATE TABLE FOOD_BOOKING(B_ID VARCHAR2(20) PRIMARY KEY,
B_DATE VARCHAR2(20) NOT NULL,MEAL_TYPE VARCHAR2(20) NOT NULL,
N_O_GUESTS VARCHAR2(20) NOT NULL,H_ID VARCHAR2(20) NOT NULL);
ALTER TABLE FOOD_BOOKING
ADD FOREIGN KEY (H_ID ) REFERENCES HOTEL_INFO (H_ID );

DESC FOOD_BOOKING;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **FOOD_BOOKING**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
FOOD_BOOKING	B_ID	Varchar2	20	-	-	1	-	-	-
	B_DATE	Varchar2	20	-	-	-	-	-	-
	MEAL_TYPE	Varchar2	20	-	-	-	-	-	-
	N_O_GUESTS	Varchar2	20	-	-	-	-	-	-
	H_ID	Varchar2	20	-	-	-	-	-	-

1 - 5

15. HOTEL_RATINGS:

```
CREATE TABLE HOTEL_RATINGS ( H_ID VARCHAR2(20) PRIMARY KEY,
RATINGS MULTIVALUE_TYPE) NESTED TABLE RATINGS
STORE AS PHONE_NUMBERS_TABLE3;

DESC HOTEL_RATINGS;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **HOTEL_RATINGS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
HOTEL_RATINGS	H_ID	Varchar2	20	-	-	1	-	-	-
	RATINGS	Multivalue_Type	16	-	-	-	✓	-	-
									1 - 2

16.PAYMENT:

```
CREATE TABLE PAYMENT(PAY_ID VARCHAR2(20) PRIMARY KEY,
AMOUNT NUMBER NOT NULL,PAY_DATE DATE NOT NULL,
PAY_METHOD VARCHAR(20) NOT NULL,INVOICE_NUMBER NUMBER NOT NULL,CUS_ID VARCHAR2(20) NOT NULL);
ALTER TABLE PAYMENT ADD FOREIGN KEY (CUS_ID) REFERENCES CUS_INFO(CUS_ID);

DESC PAYMENT;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **PAYMENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PAYMENT	PAY_ID	Varchar2	20	-	-	1	-	-	-
	AMOUNT	Number	-	-	-	-	-	-	-
	PAY_DATE	Date	7	-	-	-	-	-	-
	PAY_METHOD	Varchar2	20	-	-	-	-	-	-
	INVOICE_NUMBER	Number	-	-	-	-	-	-	-
	CUS_ID	Varchar2	20	-	-	-	-	-	-
									1 - 6

VALUE INSERTION

1. AGENCY_INFO:

Autocommit Display 10

```
INSERT ALL
INTO AGENCY_INFO VALUES('101-A','WAHIDUZZAMAN','ZAMAN@GMAIL.COM')
INTO AGENCY_INFO VALUES('102-A','DIP','DIP@GMAIL.COM')
INTO AGENCY_INFO VALUES('103-A','KHAN','KHAN@GMAIL.COM')
INTO AGENCY_INFO VALUES('104-A','KHALID','KHALID@GMAIL.COM')
INTO AGENCY_INFO VALUES('105-A','HASAN','HASAN@GMAIL.COM')
INTO AGENCY_INFO VALUES('106-A','FAYSAL','FAYSAL@GMAIL.COM')
INTO AGENCY_INFO VALUES('107-A','ESTIAK','ESTIAK@GMAIL.COM')
INTO AGENCY_INFO VALUES('108-A','AYON','AYON@GMAIL.COM')
INTO AGENCY_INFO VALUES('109-A','SHAFIN','SHAFIN@GMAIL.COM')
INTO AGENCY_INFO VALUES('101-B','JOY','JOY@GMAIL.COM')
SELECT * FROM DUAL;
SELECT * FROM AGENCY_INFO;
```

Results Explain Describe Saved SQL History

AGENCY_ID	AGENCY_NAME	EMAIL
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM
102-A	DIP	DIP@GMAIL.COM
103-A	KHAN	KHAN@GMAIL.COM
104-A	KHALID	KHALID@GMAIL.COM
105-A	HASAN	HASAN@GMAIL.COM
106-A	FAYSAL	FAYSAL@GMAIL.COM
107-A	ESTIAK	ESTIAK@GMAIL.COM
108-A	AYON	AYON@GMAIL.COM
109-A	SHAFIN	SHAFIN@GMAIL.COM
101-B	JOY	JOY@GMAIL.COM

10 rows returned in 0.00 seconds CSV Export

2. AGENCY_LOC:

Autocommit Display 10

```
INSERT ALL
INTO AGENCY_LOC VALUES ('101-A','DHAKA','BANGLADESH',1200)
INTO AGENCY_LOC VALUES ('102-A','DHAKA','BANGLADESH',1000)
INTO AGENCY_LOC VALUES ('103-A','DHAKA','BANGLADESH',1100)
INTO AGENCY_LOC VALUES ('104-A','SYLHET','BANGLADESH',3100)
INTO AGENCY_LOC VALUES ('105-A','CHATTOGRAM','BANGLADESH',4200)
INTO AGENCY_LOC VALUES ('106-A','CHATTOGRAM','BANGLADESH',4000)
INTO AGENCY_LOC VALUES ('107-A','JEDDAH','SAUDI ARABIA',22230)
INTO AGENCY_LOC VALUES ('108-A','DOHA','QATAR',122014)
INTO AGENCY_LOC VALUES ('109-A','ISTANBUL','TURKIYE',34000)
INTO AGENCY_LOC VALUES ('101-B','KUWAIT CITY','KUWAIT',13001)
SELECT * FROM DUAL;
SELECT * FROM AGENCY_LOC;
```

Results Explain Describe Saved SQL History

AGENCY_ID	CITY	COUNTRY	ZIP_CODE
101-A	DHAKA	BANGLADESH	1200
102-A	DHAKA	BANGLADESH	1000
103-A	DHAKA	BANGLADESH	1100
104-A	SYLHET	BANGLADESH	3100
105-A	CHATTOGRAM	BANGLADESH	4200
106-A	CHATTOGRAM	BANGLADESH	4000
107-A	JEDDAH	SAUDI ARABIA	22230
108-A	DOHA	QATAR	122014
109-A	ISTANBUL	TURKIYE	34000
101-B	KUWAIT CITY	KUWAIT	13001

10 rows returned in 0.02 seconds CSV Export

3. AGENCY_CON:

Autocommit Display 10

```
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('101-A',MULTIVALUE_TYPE('01711111111','01911111111'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('102-A',MULTIVALUE_TYPE('01311111111','01922222222'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('103-A',MULTIVALUE_TYPE('01612312399','01933333333'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('104-A',MULTIVALUE_TYPE('01812312399','01944444444'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('105-A',MULTIVALUE_TYPE('01512312399','01955555555'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('106-A',MULTIVALUE_TYPE('01722222222','01387654321'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('107-A',MULTIVALUE_TYPE('01733333333','01532132199'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('108-A',MULTIVALUE_TYPE('01744444444','01799999999'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('109-A',MULTIVALUE_TYPE('01755555555','01832132199'));
INSERT INTO AGENCY_CON(AGENCY_ID,PHONE_NO) VALUES('101-B',MULTIVALUE_TYPE('01766666666','01966666666'));
SELECT AGENCY_ID,COLUMN_VALUE AS PHONE_NO FROM AGENCY_CON, TABLE(AGENCY_CON.PHONE_NO);
```

Results Explain Describe Saved SQL History

AGENCY_ID	PHONE_NO
103-A	01612312399
103-A	01933333333
104-A	01812312399
104-A	01944444444
101-A	01711111111
101-A	01911111111
102-A	01311111111
102-A	01922222222
105-A	01512312399
105-A	01955555555

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds CSV Export

4. EMP_INFO:

Autocommit Display 10

```
INSERT ALL
INTO EMP_INFO VALUES ('1A','WAHIDUZZAMAN','21-JUL-95','CEO',120000,'101-A')
INTO EMP_INFO VALUES ('2A','KHALID','21-JUN-96','DIRECTOR',100000,'101-A')
INTO EMP_INFO VALUES ('3A','AYON','21-MAY-95','AS. DIRECTOR',85000,'101-A')
INTO EMP_INFO VALUES ('4A','FAYSAL','21-JUL-92','MANAGER',110000,'101-A')
INTO EMP_INFO VALUES ('1B','HASAN','21-JUL-98','CEO',150000,'102-A')
INTO EMP_INFO VALUES ('2B','KHAN','21-JAN-95','DIRECTOR',125000,'102-A')
INTO EMP_INFO VALUES ('3B','ESTIAK','21-FEB-95','MANAGER',115000,'102-A')
INTO EMP_INFO VALUES ('1C','DIP','21-MAR-95','CEO',80000,'103-A')
INTO EMP_INFO VALUES ('2C','RIDWAN','21-AUG-95','DIRECTOR',70000,'103-A')
INTO EMP_INFO VALUES ('3C','RAYYAN','21-DEC-95','MANAGER',20000,'103-A')
SELECT * FROM DUAL;
SELECT * FROM EMP_INFO;
```

Results Explain Describe Saved SQL History

E_ID	E_NAME	DOB	JOB_TITLE	SALARY	AGENCY_ID
1A	WAHIDUZZAMAN	21-JUL-95	CEO	120000	101-A
2A	KHALID	21-JUN-96	DIRECTOR	100000	101-A
3A	AYON	21-MAY-95	AS. DIRECTOR	85000	101-A
4A	FAYSAL	21-JUL-92	MANAGER	110000	101-A
1B	HASAN	21-JUL-98	CEO	150000	102-A
2B	KHAN	21-JAN-95	DIRECTOR	125000	102-A
3B	ESTIAK	21-FEB-95	MANAGER	115000	102-A
1C	DIP	21-MAR-95	CEO	80000	103-A
2C	RIDWAN	21-AUG-95	DIRECTOR	70000	103-A
3C	RAYYAN	21-DEC-95	MANAGER	20000	103-A

10 rows returned in 0.01 seconds CSV Export

5. EMP_SKILL:

Autocommit Display 10

```
INSERT ALL
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('1A',MULTIVALUE_TYPE('ORGANIZATION','PUNCTUALITY'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('2A',MULTIVALUE_TYPE('TIME MANAGEMENT','SINCERITY'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('3A',MULTIVALUE_TYPE('HONESTY','LANGUAGE SKILL'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('4A',MULTIVALUE_TYPE('PROBLEM SOLVING','INTELLECTUAL'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('1B',MULTIVALUE_TYPE('ORGANIZATION','SINCERUTY'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('2B',MULTIVALUE_TYPE('LANGUAGE SKILL','TIME MANAGEMENT'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('3B',MULTIVALUE_TYPE('HARWORKING','PUNCTUALITY'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('1C',MULTIVALUE_TYPE('HONESTY','PUNCTUALITY'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('2C',MULTIVALUE_TYPE('CODING','PROBLEM SOLVING'))
INTO EMP_SKILL(E_ID,SKILLS) VALUES ('3C',MULTIVALUE_TYPE('PROGRAMMING','INTELLECTUAL'))
SELECT * FROM DUAL;
SELECT E_ID,COLUMN_VALUE AS SKILLS FROM EMP_SKILL , TABLE(EMP_SKILL.SKILLS);
```

Results Explain Describe Saved SQL History

E_ID	SKILLS
1A	ORGANIZATION
1A	PUNCTUALITY
2A	TIME MANAGEMENT
2A	SINCERITY
3A	HONESTY
3A	LANGUAGE SKILL
4A	PROBLEM SOLVING
4A	INTELLECTUAL
1B	ORGANIZATION
1B	SINCERUTY

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds CSV Export

6. CUS_INFO:

```
INSERT ALL
INTO CUS_INFO VALUES ('10','TAMIM IQBAL','IQBAL@GMAIL.COM','DELUXE','101-A')
INTO CUS_INFO VALUES ('20','SAKIB AL HASAN','SAKIB@GMAIL.COM','PREMIUM','101-A')
INTO CUS_INFO VALUES ('30','MUSHFIQUR RAHIM','MUSHFIQ@GMAIL.COM','DELUXE','101-A')
INTO CUS_INFO VALUES ('40','MASHRAFE BIN MORTUZA','MORTUZA@GMAIL.COM','REGULAR','102-A')
INTO CUS_INFO VALUES ('50','MAHMADULLAH RIYADH','RIYADH@GMAIL.COM','PREMIUM','102-A')
INTO CUS_INFO VALUES ('60','NAZMUL HOSSAIN','SHANTO@GMAIL.COM','REGULAR','102-A')
INTO CUS_INFO VALUES ('70','TASKIN AHMED','TASKIN@GMAIL.COM','REGULAR','102-A')
INTO CUS_INFO VALUES ('80','MEHEDY HASAN','MIRAZ@GMAIL.COM','DELUXE','103-A')
INTO CUS_INFO VALUES ('90','NASUM AHMED','NASUM@GMAIL.COM','PREMIUM','103-A')
INTO CUS_INFO VALUES ('100','SHAHADAT HOSSAIN','DIPU@GMAIL.COM','REGULAR','103-A')
SELECT * FROM DUAL;
SELECT * FROM CUS_INFO;
```

Results Explain Describe Saved SQL History

CUS_ID	CUS_NAME	EMAIL	MEMBERSHIP_LEVEL	AGENCY_ID
10	TAMIM IQBAL	IQBAL@GMAIL.COM	DELUXE	101-A
20	SAKIB AL HASAN	SAKIB@GMAIL.COM	PREMIUM	101-A
30	MUSHFIQUR RAHIM	MUSHFIQ@GMAIL.COM	DELUXE	101-A
40	MASHRAFE BIN MORTUZA	MORTUZA@GMAIL.COM	REGULAR	102-A
50	MAHMADULLAH RIYADH	RIYADH@GMAIL.COM	PREMIUM	102-A
60	NAZMUL HOSSAIN	SHANTO@GMAIL.COM	REGULAR	102-A
70	TASKIN AHMED	TASKIN@GMAIL.COM	REGULAR	102-A
80	MEHEDY HASAN	MIRAZ@GMAIL.COM	DELUXE	103-A
90	NASUM AHMED	NASUM@GMAIL.COM	PREMIUM	103-A
100	SHAHADAT HOSSAIN	DIPU@GMAIL.COM	REGULAR	103-A

10 rows returned in 0.00 seconds CSV Export

7. CUS_LOC:

```
INSERT ALL
INTO CUS_LOC VALUES ('10','BANGLADESH','DHAKA','1200')
INTO CUS_LOC VALUES ('20','BANGLADESH','DHAKA','1000')
INTO CUS_LOC VALUES ('30','BANGLADESH','DHAKA','1100')
INTO CUS_LOC VALUES ('40','BANGLADESH','CHATTOGRAM','4200')
INTO CUS_LOC VALUES ('50','BANGLADESH','CHATTOGRAM','4000')
INTO CUS_LOC VALUES ('60','BANGLADESH','SYLHET','3100')
INTO CUS_LOC VALUES ('70','SAUDI ARABIA','JEDDAH','22230')
INTO CUS_LOC VALUES ('80','QATAR','DOHA','122014')
INTO CUS_LOC VALUES ('90','TURKIYE','ISTANBUL','34000')
INTO CUS_LOC VALUES ('100','KUWAIT','KUWAIT CITY','13001')
SELECT * FROM DUAL;

SELECT * FROM CUS_LOC;
```

Results Explain Describe Saved SQL History

CUS_ID	COUNTRY	CITY	ZIP_CODE
10	BANGLADESH	DHAKA	1200
20	BANGLADESH	DHAKA	1000
30	BANGLADESH	DHAKA	1100
40	BANGLADESH	CHATTOGRAM	4200
50	BANGLADESH	CHATTOGRAM	4000
60	BANGLADESH	SYLHET	3100
70	SAUDI ARABIA	JEDDAH	22230
80	QATAR	DOHA	122014
90	TURKIYE	ISTANBUL	34000
100	KUWAIT	KUWAIT CITY	13001

10 rows returned in 0.00 seconds CSV Export

8.CUS_CON:

Autocommit Display 10

```
INSERT ALL
INTO CUS_CON VALUES ('10',MULTIVALUE_TYPE('01711111111','01911111111'))
INTO CUS_CON VALUES ('20',MULTIVALUE_TYPE('01722222222','01922222222'))
INTO CUS_CON VALUES ('30',MULTIVALUE_TYPE('01733333333','01933333333'))
INTO CUS_CON VALUES ('40',MULTIVALUE_TYPE('01744444444','01944444444'))
INTO CUS_CON VALUES ('50',MULTIVALUE_TYPE('01755555555','01955555555'))
INTO CUS_CON VALUES ('60',MULTIVALUE_TYPE('01766666666','01966666666'))
INTO CUS_CON VALUES ('70',MULTIVALUE_TYPE('01788888888','01988888888'))
INTO CUS_CON VALUES ('80',MULTIVALUE_TYPE('01700000000','01900000000'))
INTO CUS_CON VALUES ('90',MULTIVALUE_TYPE('01712312399','01912312399'))
INTO CUS_CON VALUES ('100',MULTIVALUE_TYPE('01732132199','01932132199'))
SELECT * FROM DUAL;
SELECT CUS_ID,COLUMN_VALUE AS PHONE_NO FROM CUS_CON, TABLE(CUS_CON.PHONE_NO);
```

Results Explain Describe Saved SQL History

CUS_ID	PHONE_NO
10	01711111111
10	01911111111
20	01722222222
20	01922222222
30	01733333333
30	01933333333
40	01744444444
40	01944444444
50	01755555555
50	01955555555

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.02 seconds CSV Export

9.AIRLINE:

```
INSERT ALL
INTO AIRLINE VALUES ('111','QATAR AIRWAYS','10')
INTO AIRLINE VALUES ('222','FLY EMIRATES','10')
INTO AIRLINE VALUES ('333','ETIHAD AIRWAYS','10')
INTO AIRLINE VALUES ('444','QATAR AIRWAYS','20')
INTO AIRLINE VALUES ('555','ETIHAD AIRWAYS','20')
INTO AIRLINE VALUES ('666','FLY EMIRATES','20')
INTO AIRLINE VALUES ('777','BIMAN BANGLADESH','20')
INTO AIRLINE VALUES ('888','ETIHAD AIRWAYS','30')
INTO AIRLINE VALUES ('999','QATAR AIRWAYS','30')
INTO AIRLINE VALUES ('101','FLY EMIRATES','30')
SELECT * FROM DUAL;
SELECT * FROM AIRLINE;
```

Results Explain Describe Saved SQL History

FLIGHT_NO	AIRLINE	CUS_ID
111	QATAR AIRWAYS	10
222	FLY EMIRATES	10
333	ETIHAD AIRWAYS	10
444	QATAR AIRWAYS	20
555	ETIHAD AIRWAYS	20
666	FLY EMIRATES	20
777	BIMAN BANGLADESH	20
888	ETIHAD AIRWAYS	30
999	QATAR AIRWAYS	30
101	FLY EMIRATES	30

10 rows returned in 0.00 seconds CSV Export

10. AIRLINE_INFO:

```
INSERT ALL
INTO AIRLINE_INFO VALUES ('111','QATAR','BANGLADESH','10-12-23','11-12-23')
INTO AIRLINE_INFO VALUES ('222','UAE','BANGLADESH','10-12-23','11-12-23')
INTO AIRLINE_INFO VALUES ('333','UAE','BANGLADESH','15-12-23','16-12-23')
INTO AIRLINE_INFO VALUES ('444','QATAR','BANGLADESH','15-12-23','16-12-23')
INTO AIRLINE_INFO VALUES ('555','UAE','BANGLADESH','10-12-23','11-12-23')
INTO AIRLINE_INFO VALUES ('666','UAE','BANGLADESH','11-12-23','12-12-23')
INTO AIRLINE_INFO VALUES ('777','SAUDI ARABIA','BANGLADESH','10-12-23','11-12-23')
INTO AIRLINE_INFO VALUES ('888','BANGLADESH','UAE','11-12-23','12-12-23')
INTO AIRLINE_INFO VALUES ('999','BANGLADESH','QATAR','10-12-23','11-12-23')
INTO AIRLINE_INFO VALUES ('101','BANGLADESH','UAE','10-12-23','11-12-23')
SELECT * FROM DUAL;
SELECT * FROM AIRLINE_INFO;
```

Results Explain Describe Saved SQL History

FLIGHT_NO	DEP_COUNTRY	ARR_COUNTRY	DEP_TIME	ARR_TIME
111	QATAR	BANGLADESH	10-12-23	11-12-23
222	UAE	BANGLADESH	10-12-23	11-12-23
333	UAE	BANGLADESH	15-12-23	16-12-23
444	QATAR	BANGLADESH	15-12-23	16-12-23
555	UAE	BANGLADESH	10-12-23	11-12-23
666	UAE	BANGLADESH	11-12-23	12-12-23
777	SAUDI ARABIA	BANGLADESH	10-12-23	11-12-23
888	BANGLADESH	UAE	11-12-23	12-12-23
999	BANGLADESH	QATAR	10-12-23	11-12-23
101	BANGLADESH	UAE	10-12-23	11-12-23

10 rows returned in 0.00 seconds CSV Export

11. TRANSPORTATION:

```
INSERT ALL
INTO TRANSPORTATION VALUES ('111BD','11-DEC-23','DHAKA AIRPORT','PAN PACIFIC','CAR','10')
INTO TRANSPORTATION VALUES ('222BD','12-DEC-23','CHATTOGRAM AIRPORT','RADDISON BLU','CNG','10')
INTO TRANSPORTATION VALUES ('333BD','16-DEC-23','SYLHET AIRPORT','ROSE VIEW','BIKE','10')
INTO TRANSPORTATION VALUES ('444BD','17-DEC-23','DHAKA AIRPORT','PAN PACIFIC','CAR','20')
INTO TRANSPORTATION VALUES ('555BD','13-DEC-23','DHAKA AIRPORT','PAN PACIFIC','CNG','20')
INTO TRANSPORTATION VALUES ('666BD','14-DEC-23','CHATTOGRAM AIRPORT','RADDISON BLU','BIKE','20')
INTO TRANSPORTATION VALUES ('777BD','15-DEC-23','SYLHET AIRPORT','ROSE VIEW','CAR','20')
INTO TRANSPORTATION VALUES ('888BD','18-DEC-23','DUBAI AIRPORT','BURJ AL-ARAB','CAR','30')
INTO TRANSPORTATION VALUES ('999BD','19-DEC-23','QATAR AIRPORT','CROWN PLAZA','CAR','30')
INTO TRANSPORTATION VALUES ('101BD','20-DEC-23','DUBAI AIRPORT','BURJ KHALIFA','CAR','30')
SELECT * FROM DUAL;
SELECT * FROM TRANSPORTATION;
```

Results Explain Describe Saved SQL History

TRANS_NO	BOOKING_DATE	PICK_LOC	DROP_LOC	MODE_OF_TRANSPORT	CUS_ID
111BD	11-DEC-23	DHAKA AIRPORT	PAN PACIFIC	CAR	10
222BD	12-DEC-23	CHATTOGRAM AIRPORT	RADDISON BLU	CNG	10
333BD	16-DEC-23	SYLHET AIRPORT	ROSE VIEW	BIKE	10
444BD	17-DEC-23	DHAKA AIRPORT	PAN PACIFIC	CAR	20
555BD	13-DEC-23	DHAKA AIRPORT	PAN PACIFIC	CNG	20
666BD	14-DEC-23	CHATTOGRAM AIRPORT	RADDISON BLU	BIKE	20
777BD	15-DEC-23	SYLHET AIRPORT	ROSE VIEW	CAR	20
888BD	18-DEC-23	DUBAI AIRPORT	BURJ AL-ARAB	CAR	30
999BD	19-DEC-23	QATAR AIRPORT	CROWN PLAZA	CAR	30
101BD	20-DEC-23	DUBAI AIRPORT	BURJ KHALIFA	CAR	30

10 rows returned in 0.00 seconds CSV Export

12.HOTEL_INFO:

```
INSERT ALL
INTO HOTEL_INFO VALUES ('BD1','PAN PACIFIC','DHAKA','BUSINESS CENTRE','10')
INTO HOTEL_INFO VALUES ('BD2','RADDISON BLU','CHATTOGRAM','SWIMMING POOL','10')
INTO HOTEL_INFO VALUES ('BD3','ROSE VIEW','SYLHET','ROOFTOP RESTAURENT','10')
INTO HOTEL_INFO VALUES ('BD4','PAN PACIFIC','DHAKA','BUSINESS CENTRE','20')
INTO HOTEL_INFO VALUES ('BD5','PAN PACIFIC','DHAKA','BUSINESS CENTRE','20')
INTO HOTEL_INFO VALUES ('BD6','RADDISON BLU','CHATTOGRAM','SWIMMING POOL','20')
INTO HOTEL_INFO VALUES ('BD7','ROSE VIEW','SYLHET','ROOFTOP RESTAURENT','20')
INTO HOTEL_INFO VALUES ('BD8','BURJ AL-ARAB','DUBAI','DUPLEX SUITES','30')
INTO HOTEL_INFO VALUES ('BD9','CROWN PLAZA','QATAR','COMPLEMENTARY FOOD','30')
INTO HOTEL_INFO VALUES ('BD10','BURJ KHALIFA','DUBAI','DUPLEX SUITES','30')
SELECT * FROM DUAL;
SELECT * FROM HOTEL_INFO;
```

Results Explain Describe Saved SQL History

H_ID	H_NAME	LOCATION	FACILITES	CUS_ID
BD1	PAN PACIFIC	DHAKA	BUSINESS CENTRE	10
BD2	RADDISON BLU	CHATTOGRAM	SWIMMING POOL	10
BD3	ROSE VIEW	SYLHET	ROOFTOP RESTAURENT	10
BD4	PAN PACIFIC	DHAKA	BUSINESS CENTRE	20
BD5	PAN PACIFIC	DHAKA	BUSINESS CENTRE	20
BD6	RADDISON BLU	CHATTOGRAM	SWIMMING POOL	20
BD7	ROSE VIEW	SYLHET	ROOFTOP RESTAURENT	20
BD8	BURJ AL-ARAB	DUBAI	DUPLEX SUITES	30
BD9	CROWN PLAZA	QATAR	COMPLEMENTARY FOOD	30
BD10	BURJ KHALIFA	DUBAI	DUPLEX SUITES	30

10 rows returned in 0.00 seconds CSV Export

13.ACCOMODATION:

```
INSERT ALL
INTO ACCOMMODATION VALUES ('A1','11-DEC-23','15-DEC-23','PREMIUM','501','10','BD1')
INTO ACCOMMODATION VALUES ('A2','12-DEC-23','16-DEC-23','PREMIUM','301','10','BD2')
INTO ACCOMMODATION VALUES ('A3','16-DEC-23','17-DEC-23','PREMIUM','201','10','BD3')
INTO ACCOMMODATION VALUES ('A4','17-DEC-23','20-DEC-23','REGULAR','502','20','BD4')
INTO ACCOMMODATION VALUES ('A5','13-DEC-23','14-DEC-23','REGULAR','503','20','BD5')
INTO ACCOMMODATION VALUES ('A6','14-DEC-23','19-DEC-23','REGULAR','504','20','BD6')
INTO ACCOMMODATION VALUES ('A7','15-DEC-23','18-DEC-23','REGULAR','506','20','BD7')
INTO ACCOMMODATION VALUES ('A8','18-DEC-23','21-DEC-23','DELUXE','505','30','BD8')
INTO ACCOMMODATION VALUES ('A9','19-DEC-23','22-DEC-23','DELUXE','509','30','BD9')
INTO ACCOMMODATION VALUES ('A10','20-DEC-23','23-DEC-23','DELUXE','801','30','BD10')
SELECT * FROM DUAL;
SELECT * FROM ACCOMMODATION;
```

Results Explain Describe Saved SQL History

ACC_ID	IN_DATE	OUT_DATE	ROOM_TYPE	ROOM_NO	CUS_ID	H_ID
A1	11-DEC-23	15-DEC-23	PREMIUM	501	10	BD1
A2	12-DEC-23	16-DEC-23	PREMIUM	301	10	BD2
A3	16-DEC-23	17-DEC-23	PREMIUM	201	10	BD3
A4	17-DEC-23	20-DEC-23	REGULAR	502	20	BD4
A5	13-DEC-23	14-DEC-23	REGULAR	503	20	BD5
A6	14-DEC-23	19-DEC-23	REGULAR	504	20	BD6
A7	15-DEC-23	18-DEC-23	REGULAR	506	20	BD7
A8	18-DEC-23	21-DEC-23	DELUXE	505	30	BD8
A9	19-DEC-23	22-DEC-23	DELUXE	509	30	BD9
A10	20-DEC-23	23-DEC-23	DELUXE	801	30	BD10

10 rows returned in 0.01 seconds CSV Export

14. FOOD_BOOKING:

```
INSERT ALL
INTO FOOD_BOOKING VALUES ('F1','11-DEC-23','BREAKFAST','5','BD1')
INTO FOOD_BOOKING VALUES ('F2','12-DEC-23','LUNCH','3','BD1')
INTO FOOD_BOOKING VALUES ('F3','16-DEC-23','DINNER','2','BD1')
INTO FOOD_BOOKING VALUES ('F4','17-DEC-23','BREAKFAST','1','BD2')
INTO FOOD_BOOKING VALUES ('F5','13-DEC-23','BREAKFAST','5','BD2')
INTO FOOD_BOOKING VALUES ('F6','14-DEC-23','LUNCH','6','BD2')
INTO FOOD_BOOKING VALUES ('F7','15-DEC-23','LUNCH','5','BD3')
INTO FOOD_BOOKING VALUES ('F8','18-DEC-23','DINNER','4','BD3')
INTO FOOD_BOOKING VALUES ('F9','19-DEC-23','DINNER','2','BD3')
INTO FOOD_BOOKING VALUES ('F10','20-DEC-23','BREAKFAST','3','BD3')
SELECT * FROM DUAL;
SELECT * FROM FOOD_BOOKING;
```

Results Explain Describe Saved SQL History

B_ID	B_DATE	MEAL_TYPE	N_O_GUESTS	H_ID
F1	11-DEC-23	BREAKFAST	5	BD1
F2	12-DEC-23	LUNCH	3	BD1
F3	16-DEC-23	DINNER	2	BD1
F4	17-DEC-23	BREAKFAST	1	BD2
F5	13-DEC-23	BREAKFAST	5	BD2
F6	14-DEC-23	LUNCH	6	BD2
F7	15-DEC-23	LUNCH	5	BD3
F8	18-DEC-23	DINNER	4	BD3
F9	19-DEC-23	DINNER	2	BD3
F10	20-DEC-23	BREAKFAST	3	BD3

10 rows returned in 0.00 seconds CSV Export

15. HOTEL_RATINGS:

```
INSERT ALL
INTO HOTEL_RATINGS VALUES ('BD1',MULTIVALUE_TYPE('GOOD','EXCELLENT'))
INTO HOTEL_RATINGS VALUES ('BD2',MULTIVALUE_TYPE('NOT SO GOOD','AVERAGE'))
INTO HOTEL_RATINGS VALUES ('BD3',MULTIVALUE_TYPE('GOOD','AVERAGE'))
INTO HOTEL_RATINGS VALUES ('BD4',MULTIVALUE_TYPE('GOOD','EXCELLENT'))
INTO HOTEL_RATINGS VALUES ('BD5',MULTIVALUE_TYPE('GOOD','EXCELLENT'))
INTO HOTEL_RATINGS VALUES ('BD6',MULTIVALUE_TYPE('NOT SO GOOD','AVERAGE'))
INTO HOTEL_RATINGS VALUES ('BD7',MULTIVALUE_TYPE('GOOD','AVERAGE'))
INTO HOTEL_RATINGS VALUES ('BD8',MULTIVALUE_TYPE('GOOD','EXCELLENT'))
INTO HOTEL_RATINGS VALUES ('BD9',MULTIVALUE_TYPE('GOOD','EXCELLENT'))
INTO HOTEL_RATINGS VALUES ('BD10',MULTIVALUE_TYPE('GOOD','EXCELLENT'))
SELECT * FROM DUAL;
SELECT H_ID,COLUMN_VALUE AS RATINGS FROM HOTEL_RATINGS, TABLE(HOTEL_RATINGS.RATINGS);
```

Results Explain Describe Saved SQL History

H_ID	RATINGS
BD1	GOOD
BD1	EXCELLENT
BD2	NOT SO GOOD
BD2	AVERAGE
BD3	GOOD
BD3	AVERAGE
BD4	GOOD
BD4	EXCELLENT
BD5	GOOD
BD5	EXCELLENT

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.02 seconds CSV Export

16.PAYMENT:

```
INSERT ALL
INTO PAYMENT VALUES ('P1',2000,'12-DEC-23','CASH',101,'10')
INTO PAYMENT VALUES ('P2',5000,'13-DEC-23','CARD',202,'10')
INTO PAYMENT VALUES ('P3',2500,'14-DEC-23','ONLINE',303,'10')
INTO PAYMENT VALUES ('P4',2000,'12-DEC-23','CASH',404,'10')
INTO PAYMENT VALUES ('P5',5000,'13-DEC-23','CASH',505,'20')
INTO PAYMENT VALUES ('P6',6000,'14-DEC-23','CARD',606,'20')
INTO PAYMENT VALUES ('P7',4000,'15-DEC-23','ONLINE',707,'20')
INTO PAYMENT VALUES ('P8',3000,'15-DEC-23','CASH',808,'30')
INTO PAYMENT VALUES ('P9',3500,'16-DEC-23','CARD',909,'30')
INTO PAYMENT VALUES ('P10',6500,'18-DEC-23','ONLINE',999,'30')
SELECT * FROM DUAL;
SELECT * FROM PAYMENT;
```

Results Explain Describe Saved SQL History

PAY_ID	AMOUNT	PAY_DATE	PAY_METHOD	INVOICE_NUMBER	CUS_ID
P1	2000	12-DEC-23	CASH	101	10
P2	5000	13-DEC-23	CARD	202	10
P3	2500	14-DEC-23	ONLINE	303	10
P4	2000	12-DEC-23	CASH	404	10
P5	5000	13-DEC-23	CASH	505	20
P6	6000	14-DEC-23	CARD	606	20
P7	4000	15-DEC-23	ONLINE	707	20
P8	3000	15-DEC-23	CASH	808	30
P9	3500	16-DEC-23	CARD	909	30
P10	6500	18-DEC-23	ONLINE	999	30

10 rows returned in 0.00 seconds CSV Export

Query Testing

Single Row Query

Q-01:Show all information of Agency which ID ends with ‘A’.

```
SELECT * FROM AGENCY_INFO
WHERE AGENCY_ID LIKE '%A';
```

Results Explain Describe Saved SQL History

AGENCY_ID	AGENCY_NAME	EMAIL
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM
102-A	DIP	DIP@GMAIL.COM
103-A	KHAN	KHAN@GMAIL.COM
104-A	KHALID	KHALID@GMAIL.COM
105-A	HASAN	HASAN@GMAIL.COM
106-A	FAYSAL	FAYSAL@GMAIL.COM
107-A	ESTIAK	ESTIAK@GMAIL.COM
108-A	AYON	AYON@GMAIL.COM
109-A	SHAFIN	SHAFIN@GMAIL.COM

9 rows returned in 0.00 seconds CSV Export

Q-02:Show customer’s payment ranges from 3000 to 6000.

Autocommit Display 10

```
SELECT * FROM PAYMENT
WHERE AMOUNT BETWEEN 3000 AND 6000;
```

Results Explain Describe Saved SQL History

PAY_ID	AMOUNT	PAY_DATE	PAY_METHOD	INVOICE_NUMBER	CUS_ID
P2	5000	13-DEC-23	CARD	202	10
P5	5000	13-DEC-23	CASH	505	20
P6	6000	14-DEC-23	CARD	606	20
P7	4000	15-DEC-23	ONLINE	707	20
P8	3000	15-DEC-23	CASH	808	30
P9	3500	16-DEC-23	CARD	909	30

6 rows returned in 0.02 seconds CSV Export

Q-03:Show the list of customers who have Deluxe,Premium membership.

Autocommit Display 10

```
SELECT * FROM CUS_INFO
WHERE MEMBERSHIP_LEVEL IN('DELUXE','PREMIUM');
```

Results Explain Describe Saved SQL History

CUS_ID	CUS_NAME	EMAIL	MEMBERSHIP_LEVEL	AGENCY_ID
10	TAMIM IQBAL	IQBAL@GMAIL.COM	DELUXE	101-A
20	SAKIB AL HASAN	SAKIB@GMAIL.COM	PREMIUM	101-A
30	MUSHFIQUR RAHIM	MUSHFIQ@GMAIL.COM	DELUXE	101-A
50	MAHMADULLAH RIYADH	RIYADH@GMAIL.COM	PREMIUM	102-A
80	MEHEDY HASAN	MIRAZ@GMAIL.COM	DELUXE	103-A
90	NASUM AHMED	NASUM@GMAIL.COM	PREMIUM	103-A

6 rows returned in 0.02 seconds CSV Export

Multiple Row Query

Q-04.Show the salaries of CEOs who earn more than 120,000.

Autocommit Display 10

```
SELECT * FROM EMP_INFO
WHERE JOB_TITLE='CEO'
AND SALARY >=120000;
```

Results Explain Describe Saved SQL History

E_ID	E_NAME	DOB	JOB_TITLE	SALARY	AGENCY_ID
1A	WAHIDUZZAMAN	21-JUL-95	CEO	120000	101-A
1B	HASAN	21-JUL-98	CEO	150000	102-A

2 rows returned in 0.00 seconds CSV Export

Q-05.Show the information of employees who are CEO or earn more than 120,000.

Autocommit Display 10

```
SELECT * FROM EMP_INFO
WHERE JOB_TITLE='CEO'
OR SALARY >=120000;
```

Results Explain Describe Saved SQL History

E_ID	E_NAME	DOB	JOB_TITLE	SALARY	AGENCY_ID
1A	WAHIDUZZAMAN	21-JUL-95	CEO	120000	101-A
1B	HASAN	21-JUL-98	CEO	150000	102-A
2B	KHAN	21-JAN-95	DIRECTOR	125000	102-A
1C	DIP	21-MAR-95	CEO	80000	103-A

4 rows returned in 0.02 seconds CSV Export

Q-06.Show the information of the employees who are not CEO.Also create a temporary column and name it “Annual Salary” and arrange the table by descending order based on salary.

Autocommit Display 10

```
SELECT E_NAME,JOB_TITLE,SALARY*12 "ANNUAL SALARY" FROM EMP_INFO
WHERE JOB_TITLE<>'CEO'
ORDER BY SALARY DESC;
```

Results Explain Describe Saved SQL History

E_NAME	JOB_TITLE	ANNUAL SALARY
KHAN	DIRECTOR	1500000
ESTIAK	MANAGER	1380000
FAYSAL	MANAGER	1320000
KHALID	DIRECTOR	1200000
AYON	AS. DIRECTOR	1020000
RIDWAN	DIRECTOR	840000
RAYYAN	MANAGER	240000

7 rows returned in 0.02 seconds CSV Export

Sub-query(Single Row)

Q-07.Show the customers information who owns same membership as of "TAMIM IQBAL".

Autocommit Display 10

```
SELECT * FROM CUS_INFO
WHERE MEMBERSHIP_LEVEL = (
SELECT MEMBERSHIP_LEVEL FROM CUS_INFO
WHERE CUS_NAME='TAMIM IQBAL');
```

Results Explain Describe Saved SQL History

CUS_ID	CUS_NAME	EMAIL	MEMBERSHIP_LEVEL	AGENCY_ID
10	TAMIM IQBAL	IQBAL@GMAIL.COM	DELUXE	101-A
30	MUSHFIQUR RAHIM	MUSHFIQ@GMAIL.COM	DELUXE	101-A
80	MEHEDY HASAN	MIRAZ@GMAIL.COM	DELUXE	103-A

3 rows returned in 0.01 seconds CSV Export

Q-08.Show the information of employees who are at the same position as of "WAHIDUZZAMAN" and earns more than "KHALID".

Autocommit Display 10

```
SELECT * FROM EMP_INFO
WHERE JOB_TITLE=( SELECT JOB_TITLE
FROM EMP_INFO WHERE E_NAME='WAHIDUZZAMAN')
AND SALARY > (SELECT SALARY
FROM EMP_INFO WHERE E_NAME='KHALID');
```

Results Explain Describe Saved SQL History

E_ID	E_NAME	DOB	JOB_TITLE	SALARY	AGENCY_ID
1A	WAHIDUZZAMAN	21-JUL-95	CEO	120000	101-A
1B	HASAN	21-JUL-98	CEO	150000	102-A

2 rows returned in 0.00 seconds CSV Export

Sub-Query (Multiple Row)

Q-09.Show Payment information more than any of the customer id 10.

Autocommit Display 10

```
SELECT * FROM PAYMENT
WHERE AMOUNT > ANY (
SELECT AMOUNT FROM PAYMENT
WHERE CUS_ID=10);
```

Results Explain Describe Saved SQL History

PAY_ID	AMOUNT	PAY_DATE	PAY_METHOD	INVOICE_NUMBER	CUS_ID
P2	5000	13-DEC-23	CARD	202	10
P3	2500	14-DEC-23	ONLINE	303	10
P5	5000	13-DEC-23	CASH	505	20
P6	6000	14-DEC-23	CARD	606	20
P7	4000	15-DEC-23	ONLINE	707	20
P8	3000	15-DEC-23	CASH	808	30
P9	3500	16-DEC-23	CARD	909	30
P10	6500	18-DEC-23	ONLINE	999	30

8 rows returned in 0.00 seconds CSV Export

Q-10.Show Payment information more than all the customer id 10.

Autocommit Display 10

```
SELECT * FROM PAYMENT
WHERE AMOUNT > ALL (
SELECT AMOUNT FROM PAYMENT
WHERE CUS_ID=10);
```

Results Explain Describe Saved SQL History

PAY_ID	AMOUNT	PAY_DATE	PAY_METHOD	INVOICE_NUMBER	CUS_ID
P6	6000	14-DEC-23	CARD	606	20
P10	6500	18-DEC-23	ONLINE	999	30

2 rows returned in 0.00 seconds CSV Export

Q-11.Show the list of Payments which are as same as customer id 10.

Autocommit Display 10

```
SELECT * FROM PAYMENT
WHERE AMOUNT IN (
SELECT AMOUNT FROM PAYMENT
WHERE CUS_ID=10);
```

Results Explain Describe Saved SQL History

PAY_ID	AMOUNT	PAY_DATE	PAY_METHOD	INVOICE_NUMBER	CUS_ID
P4	2000	12-DEC-23	CASH	404	10
P1	2000	12-DEC-23	CASH	101	10
P5	5000	13-DEC-23	CASH	505	20
P2	5000	13-DEC-23	CARD	202	10
P3	2500	14-DEC-23	ONLINE	303	10

5 rows returned in 0.00 seconds CSV Export

Aggregate Function Query

Q-12.Show maximum and minimum salary from Emp_Info table using group by clause.

Autocommit Display 10

```
SELECT JOB_TITLE,MAX(SALARY),MIN(SALARY)
FROM EMP_INFO
GROUP BY JOB_TITLE;
```

Results Explain Describe Saved SQL History

JOB_TITLE	MAX(SALARY)	MIN(SALARY)
AS. DIRECTOR	85000	85000
MANAGER	115000	20000
CEO	150000	80000
DIRECTOR	125000	70000

4 rows returned in 0.00 seconds CSV Export

Q-13.Show maximum,minimum,average and sum of the salaries of the managers from Emp_info Table.

Autocommit Display 10

```
SELECT MAX(SALARY),MIN(SALARY),SUM(SALARY),ROUND(AVG(SALARY),0)
FROM EMP_INFO
WHERE JOB_TITLE='MANAGER';
```

Results Explain Describe Saved SQL History

MAX(SALARY)	MIN(SALARY)	SUM(SALARY)	ROUND(AVG(SALARY),0)
115000	20000	245000	81667

1 rows returned in 0.01 seconds CSV Export

Q-14.Show average amounts of payment which is more than 3000 using group by clause.

Autocommit Display 10

```
SELECT CUS_ID,ROUND(AVG(AMOUNT),0)
FROM PAYMENT GROUP BY CUS_ID
HAVING AVG(AMOUNT)>3000;
```

Results Explain Describe Saved SQL History

CUS_ID	ROUND(AVG(AMOUNT),0)
20	5000
30	4333

2 rows returned in 0.00 seconds CSV Export

Q-15.Show the maximum average salary from Emp_Info Table.

Autocommit Display 10

```
SELECT MAX(AVG(SALARY)) FROM EMP_INFO
GROUP BY E_ID;
```

Results Explain Describe Saved SQL History

MAX(AVG(SALARY))
150000

1 rows returned in 0.00 seconds CSV Export

Q-16.Show the number of hotels situated in Dhaka.

Autocommit Display 10

```
SELECT COUNT(*) FROM HOTEL_INFO
WHERE LOCATION='DHAKA';
```

Results Explain Describe Saved SQL History

COUNT(*)
3

1 rows returned in 0.00 seconds CSV Export

Joining Query

Q-17.Show all the payments information of all the customers. (Equi)

Autocommit Display 10

```
SELECT C.CUS_NAME,P.*
FROM CUS_INFO C,PAYMENT P
WHERE C.CUS_ID=P.CUS_ID;
```

Results Explain Describe Saved SQL History

CUS_NAME	PAY_ID	AMOUNT	PAY_DATE	PAY_METHOD	INVOICE_NUMBER	CUS_ID
TAMIM IQBAL	P1	2000	12-DEC-23	CASH	101	10
TAMIM IQBAL	P2	5000	13-DEC-23	CARD	202	10
TAMIM IQBAL	P3	2500	14-DEC-23	ONLINE	303	10
TAMIM IQBAL	P4	2000	12-DEC-23	CASH	404	10
SAKIB AL HASAN	P5	5000	13-DEC-23	CASH	505	20
SAKIB AL HASAN	P6	6000	14-DEC-23	CARD	606	20
SAKIB AL HASAN	P7	4000	15-DEC-23	ONLINE	707	20
MUSHFIQUR RAHIM	P8	3000	15-DEC-23	CASH	808	30
MUSHFIQUR RAHIM	P9	3500	16-DEC-23	CARD	909	30
MUSHFIQUR RAHIM	P10	6500	18-DEC-23	ONLINE	999	30

10 rows returned in 0.02 seconds CSV Export

Q-18.Show flights and hotels information together of the customers.(Equi)

Autocommit Display 10

```
SELECT AIR.*,H.*
FROM AIRLINE AIR,HOTEL_INFO H
WHERE AIR.CUS_ID=H.CUS_ID;
```

Results Explain Describe Saved SQL History

FLIGHT_NO	AIRLINE	CUS_ID	H_ID	H_NAME	LOCATION	FACILITES	CUS_ID
333	ETIHAD AIRWAYS	10	BD1	PAN PACIFIC	DHAKA	BUSINESS CENTRE	10
222	FLY EMIRATES	10	BD1	PAN PACIFIC	DHAKA	BUSINESS CENTRE	10
111	QATAR AIRWAYS	10	BD1	PAN PACIFIC	DHAKA	BUSINESS CENTRE	10
333	ETIHAD AIRWAYS	10	BD2	RADDISON BLU	CHATTOGRAM	SWIMMING POOL	10
222	FLY EMIRATES	10	BD2	RADDISON BLU	CHATTOGRAM	SWIMMING POOL	10
111	QATAR AIRWAYS	10	BD2	RADDISON BLU	CHATTOGRAM	SWIMMING POOL	10
333	ETIHAD AIRWAYS	10	BD3	ROSE VIEW	SYLHET	ROOFTOP RESTAURENT	10
222	FLY EMIRATES	10	BD3	ROSE VIEW	SYLHET	ROOFTOP RESTAURENT	10
111	QATAR AIRWAYS	10	BD3	ROSE VIEW	SYLHET	ROOFTOP RESTAURENT	10
777	BIMAN BANGLADESH	20	BD4	PAN PACIFIC	DHAKA	BUSINESS CENTRE	20

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds CSV Export

Q-19.Show the agencial information of all the employees.(Equi-joins)

Autocommit Display 10

```
SELECT A.*,E.*
FROM AGENCY_INFO A,EMP_INFO E
WHERE A.AGENCY_ID=E.AGENCY_ID;
```

Results Explain Describe Saved SQL History

AGENCY_ID	AGENCY_NAME	EMAIL	E_ID	E_NAME	DOB	JOB_TITLE	SALARY	AGENCY_ID
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	1A	WAHIDUZZAMAN	21-JUL-95	CEO	120000	101-A
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	2A	KHALID	21-JUN-96	DIRECTOR	100000	101-A
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	3A	AYON	21-MAY-95	AS. DIRECTOR	85000	101-A
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	4A	FAYSAL	21-JUL-92	MANAGER	110000	101-A
102-A	DIP	DIP@GMAIL.COM	1B	HASAN	21-JUL-98	CEO	150000	102-A
102-A	DIP	DIP@GMAIL.COM	2B	KHAN	21-JAN-95	DIRECTOR	125000	102-A
102-A	DIP	DIP@GMAIL.COM	3B	ESTIAK	21-FEB-95	MANAGER	115000	102-A
103-A	KHAN	KHAN@GMAIL.COM	1C	DIP	21-MAR-95	CEO	80000	103-A
103-A	KHAN	KHAN@GMAIL.COM	2C	RIDWAN	21-AUG-95	DIRECTOR	70000	103-A
103-A	KHAN	KHAN@GMAIL.COM	3C	RAYYAN	21-DEC-95	MANAGER	20000	103-A

10 rows returned in 0.01 seconds CSV Export

Q-20.Get all the employees who works in the same position as of 'WAHIDUZZAMAN'.(self-joins)

Autocommit Display 10

```
SELECT A.*
FROM EMP_INFO A, EMP_INFO B
WHERE A.JOB_TITLE=B.JOB_TITLE
AND B.E_NAME='WAHIDUZZAMAN'
AND A.E_NAME<>'WAHIDUZZAMAN';
```

Results Explain Describe Saved SQL History

E_ID	E_NAME	DOB	JOB_TITLE	SALARY	AGENCY_ID
1B	HASAN	21-JUL-98	CEO	150000	102-A
1C	DIP	21-MAR-95	CEO	80000	103-A

2 rows returned in 0.07 seconds CSV Export

Q-21.Get all the matching records from Agency and Employee table.

(Full outer-joins)

Autocommit Display 10

```
SELECT AGENCY_INFO.*,EMP_INFO.*
FROM AGENCY_INFO FULL OUTER JOIN EMP_INFO
ON AGENCY_INFO.AGENCY_ID=EMP_INFO.AGENCY_ID;
```

Results Explain Describe Saved SQL History

AGENCY_ID	AGENCY_NAME	EMAIL	E_ID	E_NAME	DOB	JOB_TITLE	SALARY	AGENCY_ID
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	1A	WAHIDUZZAMAN	21-JUL-95	CEO	120000	101-A
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	2A	KHALID	21-JUN-96	DIRECTOR	100000	101-A
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	3A	AYON	21-MAY-95	AS. DIRECTOR	85000	101-A
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM	4A	FAYSAL	21-JUL-92	MANAGER	110000	101-A
102-A	DIP	DIP@GMAIL.COM	1B	HASAN	21-JUL-98	CEO	150000	102-A
102-A	DIP	DIP@GMAIL.COM	2B	KHAN	21-JAN-95	DIRECTOR	125000	102-A
102-A	DIP	DIP@GMAIL.COM	3B	ESTIAK	21-FEB-95	MANAGER	115000	102-A
103-A	KHAN	KHAN@GMAIL.COM	1C	DIP	21-MAR-95	CEO	80000	103-A
103-A	KHAN	KHAN@GMAIL.COM	2C	RIDWAN	21-AUG-95	DIRECTOR	70000	103-A
103-A	KHAN	KHAN@GMAIL.COM	3C	RAYYAN	21-DEC-95	MANAGER	20000	103-A

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.02 seconds CSV Export

Q-22. Display all the agencial information for all the existing employees using left outer join.

Autocommit Display 10

```
SELECT AGENCY.AGENCY_ID, AGENCY.AGENCY_NAME,EMP.E_ID,EMP.E_NAME
FROM AGENCY_INFO AGENCY LEFT OUTER JOIN EMP_INFO EMP
ON AGENCY.AGENCY_ID=EMP.AGENCY_ID;
```

Results Explain Describe Saved SQL History

AGENCY_ID	AGENCY_NAME	E_ID	E_NAME
101-A	WAHIDUZZAMAN	1A	WAHIDUZZAMAN
101-A	WAHIDUZZAMAN	2A	KHALID
101-A	WAHIDUZZAMAN	3A	AYON
101-A	WAHIDUZZAMAN	4A	FAYSAL
102-A	DIP	1B	HASAN
102-A	DIP	2B	KHAN
102-A	DIP	3B	ESTIAK
103-A	KHAN	1C	DIP
103-A	KHAN	2C	RIDWAN
103-A	KHAN	3C	RAYYAN

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds CSV Export

Q-23. Display all the agencial information for all the existing employees using right outer join.

Autocommit Display 10

```
SELECT AGENCY.AGENCY_ID, AGENCY.AGENCY_NAME,EMP.E_ID,EMP.E_NAME
FROM EMP_INFO EMP RIGHT OUTER JOIN AGENCY_INFO AGENCY
ON AGENCY.AGENCY_ID=EMP.AGENCY_ID;
```

Results Explain Describe Saved SQL History

AGENCY_ID	AGENCY_NAME	E_ID	E_NAME
101-A	WAHIDUZZAMAN	1A	WAHIDUZZAMAN
101-A	WAHIDUZZAMAN	2A	KHALID
101-A	WAHIDUZZAMAN	3A	AYON
101-A	WAHIDUZZAMAN	4A	FAYSAL
102-A	DIP	1B	HASAN
102-A	DIP	2B	KHAN
102-A	DIP	3B	ESTIAK
103-A	KHAN	1C	DIP
103-A	KHAN	2C	RIDWAN
103-A	KHAN	3C	RAYYAN

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds CSV Export

View Query

Q-24. Create a view called Airways based on flight_no,Airline and customer id where customer id is 10.

```
CREATE VIEW AIRWAYS AS
SELECT FLIGHT_NO,AIRLINE,CUS_ID
FROM AIRLINE WHERE CUS_ID=10;
DESC AIRWAYS;
```

Results Explain Describe Saved SQL History

Object Type **VIEW** Object **AIRWAYS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
AIRWAYS	FLIGHT_NO	Varchar2	20	-	-	-	-	-	-
	AIRLINE	Varchar2	20	-	-	-	-	-	-
	CUS_ID	Varchar2	20	-	-	-	-	-	-

1 - 3

Q-25. Create a view called Housing based on types of room,room no,check in and check out dates.

Autocommit Display 10

```
CREATE VIEW HOUSING AS
SELECT ROOM_TYPE,ROOM_NO,IN_DATE,OUT_DATE
FROM ACCOMMODATION;
DESC HOUSING;
```

Results Explain Describe Saved SQL History

Object Type **VIEW** Object **HOUSING**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
HOUSING	ROOM_TYPE	Varchar2	20	-	-	-	✓	-	-
	ROOM_NO	Number	-	7	0	-	✓	-	-
	IN_DATE	Date	7	-	-	-	-	-	-
	OUT_DATE	Date	7	-	-	-	-	-	-

1 - 4

Q-26. Create a view “Transaction” that contains Pay_ID, Amount, Pay_method , Cus_ID. Level the columns respectively as Payment_ID,Value, Pay_Way, Customer_ID. Make Sure that no DML operation can be performed on this view.

Autocommit Display 10

```
CREATE VIEW TRANSACTIONS AS
SELECT PAY_ID AS PAYMENT_ID,AMOUNT AS VALUE,
PAY_METHOD AS PAY_WAY,CUS_ID AS CUSTOMER_ID
FROM PAYMENT WHERE CUS_ID=10
WITH READ ONLY;

DESC TRANSACTIONS;
```

Results Explain Describe Saved SQL History

Object Type **VIEW** Object **TRANSACTIONS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TRANSACTIONS	PAYMENT_ID	Varchar2	20	-	-	-	-	-	-
	VALUE	Number	-	-	-	-	-	-	-
	PAY_WAY	Varchar2	20	-	-	-	-	-	-
	CUSTOMER_ID	Varchar2	20	-	-	-	-	-	-

1 - 4

Database Connection

Process:

To connect a Database with JDBC we need 3 things.

1. MySQL java connector (jar file). You can go to (Maven Repository: mysql » mysql-connector-java (mvnrepository.com)) and download any version you like.
2. We need XAMPP. Xampp can be downloaded from here (Download XAMPP (apachefriends.org)).
3. We can use any IDE. In our project netbeans has been used.

A. Start the XAMPP app then start the both Apache and MySQL. Next go to MySQL admin panel.
B. Now create a Database, inside of it create a table and insert the values in the table.

AGENCY_ID	AGENCY_NAME	EMAIL
101-A	WAHIDUZZAMAN	ZAMAN@GMAIL.COM
102-A	DIP	DIP@GMAIL.COM
103-A	KHAN	KHAN@GMAIL.COM
104-A	KHALID	KHALID@GMAIL.COM
105-A	HASAN	HASAN@GMAIL.COM
106-A	FAYSAL	FAYSAL@GMAIL.COM
107-A	ESTIAK	ESTIAK@GMAIL.COM
108-A	AYON	AYON@GMAIL.COM
109-A	SHAFIN	SHAFIN@GMAIL.COM
101-B	JOY	JOY@GMAIL.COM

Show all | Number of rows: 25 Filter rows: Search this table

C. Next we opened the IDE (Netbeans).

i. Create a project. Add the jar file to the libraries.

ii. In source package create folder.In the folder create a java file.

iii. Now to write the code we need to understand the steps of Database connection.

- Register driver.
- Connect the database.
- Create a statement.
- Execute the query().
- Connection close().

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package JDBC;
import java.sql.*;
public class JDBCTEST {
    public static void main (String[]args) throws ClassNotFoundException, SQLException
    {
        Class.forName("com.mysql.cj.jdbc.Driver");
        String user = "root";
        String password ="";
        String url = "jdbc:mysql://127.0.0.1:3306/TOURISM_MANAGEMENT?serverTimezone=UTC";

        Connection conn = DriverManager.getConnection(url, user, password);
        Statement statement = conn.createStatement();
        String query ="SELECT * FROM AGENCY_INFO";
        ResultSet result = statement.executeQuery(query);
        while(result.next())
        {
            String AGENCY_ID,AGENCY_NAME,EMAIL;
            AGENCY_ID = result.getString("AGENCY_ID");
            AGENCY_NAME = result.getString("AGENCY_NAME");
            EMAIL = result.getString("EMAIL");

            System.out.println(AGENCY_ID+" "+AGENCY_NAME+" "+EMAIL);


        }
        statement.close();
        conn.close();
    }
}
```

Fig: Connection Code

D. The output is:

```
Output - TOURISM_MANAGEMENT (run)
run:
101-A WAHIDUZZAMAN ZAMAN@GMAIL.COM
102-A DIP DIP@GMAIL.COM
103-A KHAN KHAN@GMAIL.COM
104-A KHALID KHALID@GMAIL.COM
105-A HASAN HASAN@GMAIL.COM
106-A FAYSAL FAYSAL@GMAIL.COM
107-A ESTIAK ESTIAK@GMAIL.COM
108-A AYON AYON@GMAIL.COM
109-A SHAFIN SHAFIN@GMAIL.COM
101-B JOY JOY@GMAIL.COM
BUILD SUCCESSFUL (total time: 0 seconds)
```

E. Finished.

About Database Connection & Its Features

A database connection is like a virtual bridge between a computer program (like a website or application) and a storage area for information, known as a database. It allows the program to interact with and retrieve data from the database.

Key Features:

1. **Connection String:**
    - Think of it as a secret code that the program uses to find the database. It contains details like where the database is located and how to get in.
2. **Connection Pooling:**
    - Imagine a shared pool of workers. Instead of hiring new workers every time a job needs to be done, the program reuses existing workers to save time and effort.
3. **Authentication:**
    - Similar to showing a badge to enter a secure area, authentication ensures that only authorized users or programs can access the database.
4. **Transaction Management:**
    - Transactions are like making several moves in a game. Either all the moves happen together, or none of them do. This ensures that the database stays in a good state.
5. **Error Handling:**
    - When something goes wrong (like a misspelled command), error handling helps the program understand and respond appropriately without crashing.
6. **Concurrency Control:**
    - Imagine people sharing a document. Concurrency control ensures that changes made by one person don't mess up what another person is doing.
7. **Database Drivers:**
    - Just as you need a specific key for different locks, database drivers are like keys that help the program communicate with different types of databases.
8. **SSL/TLS Encryption:**
    - It's like sending a secret letter in a locked box. SSL/TLS encryption ensures that information traveling between the program and the database is secure and private.

9. **Connection Timeouts:**
    - This is like setting a timer. If the program can't connect to the database within a certain time, it gives up to prevent waiting forever.
10. **Connection States:**
    - Think of a connection as a light switch—it can be off, on, or doing something. Understanding connection states helps the program manage its interactions with the database effectively.

In simple terms, database connection features are like a set of rules and tools that help programs talk to databases in a secure, reliable, and efficient way.

Conclusion

Our tourist management system project is designed to work as a database schema for any travel management system. We made an effort to incorporate all entity kinds and store their data in the most straightforward manner imaginable. **Through this project, we have applied the theoretical knowledge we learned in this Introduction to Database course to a real-world project. The information we have gained from this project and the training will hopefully be useful in developing any database-related projects we work on in the future.** We are also very excited to work on this project and willing to add more features as well.